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Reasoning Under Attack: A Review of Neuro-Symbolic AI for Explainable and Trustworthy Cybersecurity

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Abstract:	The influence of dynamic, increasingly sophisticated cyberattacks, combined with the rigidity of either fully neural or purely symbolic AI, has created a clear and pressing need for hybrid solutions that are both pattern-aware and principle-aware. Neuro-Symbolic AI (NSAI), defined as the convergence of the learning ability of deep neural networks and the reasoning engines of symbolic AI, has emerged as one potentially compelling solution but lacks a comprehensive, cross-domain synthesis of its applications, evaluation methodologies and open problems in the cybersecurity domain. The review presented here fills this need, analyzing in detail 127 peer-reviewed papers retrieved from Springer, IEEE Xplore, ACM Digital Library and Scopus over the period 2019 - 2025. Through the lens of VOSviewer's keyword co-occurrence, we identify six major clusters across 312 indexed papers: threat detection, knowledge representation, IoT security, trustworthiness, introductory and overview literature, and emerging frontiers. We elaborate on these clusters and the inherent design limitations of the current field before providing detailed descriptions of the core tenets underlying existing NSAI frameworks such as Logic Tensor Networks, DeepProbLog, Neural Theorem Provers, and the Neuro-Symbolic Concept Learner, then assessing NSAI applications in six security-relevant domains. Overall, for metrics critical for real-world applications such as zero-day detection accuracy, false-positive rates, and explainability metrics, NSAI approaches outperform pure deep learning baselines by 20-50%. In a comparative analysis of both pure deep learning and purely rule-based expert systems, NSAI performs superior across explainability, adversarial robustness, regulatory compliance, and knowledge integration; raw accuracy and inference speed continue to favour pure deep learning approaches. Major challenges facing the field revolve around the integration of symbolic reasoning in high-throughput, real-time SOC scenarios, the problem of knowledge acquisition in dynamic security environments, existing SOC infrastructure integration issues, and the governance and regulatory challenges of dual-use autonomous cyber defence agents. Finally, we highlight Causal Reasoning and Large Language Model convergence as the two most exciting and promising near-term frontiers and propose an organized roadmap focused on three core directions: grounding, instructibility and alignment.
Opposed Reviewers:	

From

09/06/2026

Dr. Shanavaz H

Assistant Professor

Department of Chemistry and Bio-Chemistry

Jain (Deemed-to-be University)

To,

Editor-in-Chief,

Artificial Intelligence Journal

Subject: Submission of Review Article — "Reasoning Under Attack: A Review of Neuro-Symbolic AI for Explainable and Trustworthy Cybersecurity"

Dear Professor Furnell,

I am pleased to submit our review article, "Reasoning Under Attack: A Review of Neuro-Symbolic AI for Explainable and Trustworthy Cybersecurity," for consideration for publication in *Computers & Security*. This manuscript is original, has not been published previously, and is not under consideration elsewhere. The review provides the first systematic, cybersecurity-focused synthesis of neuro-symbolic AI (NSAI), addressing the growing need for approaches that combine the pattern-recognition capabilities of neural networks with the logical reasoning strengths of symbolic AI. The manuscript includes a bibliometric analysis of 312 peer-reviewed publications (2019–2025), evaluates major NSAI frameworks across six cybersecurity domains, presents a comparative assessment against deep learning and rule-based systems across fourteen performance dimensions, and identifies key research gaps while proposing a future research roadmap based on the Grounding–Instructibility–Alignment (G-I-A) framework.

The manuscript aligns closely with the scope of *Computers & Security* by addressing AI-driven threat detection, malware analysis, intrusion detection, adversarial robustness, and explainable cybersecurity. Given the increasing demands for transparency and accountability under regulations such as GDPR and NIS2, the explainability and auditability offered by NSAI systems are particularly relevant to the journal's readership. The manuscript comprises approximately 10,000 words, 15 figures, 5 tables, and 87 references. All authors have approved this submission, the work is original, and there are no conflicts of interest to declare. We appreciate your consideration and would be pleased to suggest potential reviewers if required.

Yours faithfully,

Dr. Shanavaz H

Reasoning Under Attack: A Review of Neuro-Symbolic AI for Explainable and Trustworthy Cybersecurity

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Abstract:

The influence of dynamic, increasingly sophisticated cyberattacks, combined with the rigidity of either fully neural or purely symbolic AI, has created a clear and pressing need for hybrid solutions that are both pattern-aware and principle-aware. Neuro-Symbolic AI (NSAI), defined as the convergence of the learning ability of deep neural networks and the reasoning engines of symbolic AI, has emerged as one potentially compelling solution but lacks a comprehensive, cross-domain synthesis of its applications, evaluation methodologies and open problems in the cybersecurity domain. The review presented here fills this need, analyzing in detail 127 peer-reviewed papers retrieved from Springer, IEEE Xplore, ACM Digital Library and Scopus over the period 2019 - 2025. Through the lens of VOSviewer's keyword co-occurrence, we identify six major clusters across 312 indexed papers: threat detection, knowledge representation, IoT security, trustworthiness, introductory and overview literature, and emerging frontiers. We elaborate on these clusters and the inherent design limitations of the current field before providing detailed descriptions of the core tenets underlying existing NSAI frameworks such as Logic Tensor Networks, DeepProbLog, Neural Theorem Provers, and the Neuro-Symbolic Concept Learner, then assessing NSAI applications in six security-relevant domains. Overall, for metrics critical for real-world applications such as zero-day detection accuracy, false-positive rates, and explainability metrics, NSAI approaches outperform pure deep learning baselines by 20-50%. In a comparative analysis of both pure deep learning and purely rule-based expert systems, NSAI performs superior across explainability, adversarial robustness, regulatory compliance, and knowledge integration; raw accuracy and inference speed continue to favour pure deep learning approaches. Major challenges facing the field revolve around the integration of symbolic reasoning in high-throughput, real-time SOC scenarios, the problem of knowledge acquisition in dynamic security environments, existing SOC infrastructure integration issues, and the governance and regulatory challenges of dual-use autonomous cyber defence agents. Finally, we highlight Causal Reasoning and Large Language Model convergence as the two most exciting and promising near-term frontiers and propose an organized roadmap focused on three core directions: grounding, instructibility and alignment.

Key words: Neuro-symbolic AI, Cybersecurity, Intrusion detection, Threat intelligence, Symbolic reasoning.

1. Introduction

The field of AI has traditionally existed in a boom-and-bust cycle. Intense periods of research, optimism, and widespread acceptance known as "AI summers" are interspersed with lulls, during which research efforts slow. Interest declined, known as "AI winters," and we find ourselves today (Kautz, 2022). One of the more consequential outcomes of this period is the merging of two historically separate fields: Symbolic AI and Sub-Symbolic AI, a combination known as Neuro-Symbolic AI. Whether this fusion is actually necessary remains debated; some argue that common-sense reasoning can be solved with enough data (A Path Towards Autonomous Machine Intelligence - Version 0.9.2, 2022-06-27 - Yann LeCun | Arxiv, n.d.), while others push back sharply — one memorable objection being that "You can't get to the moon by climbing successively taller trees" (Marcus and Davis, 2019). This paper takes the position that symbolic reasoning is indispensable, and that Neuro-Symbolic AI is the most productive direction for the field. Accordingly, it presents a systematic literature review of prominent Neuro-Symbolic projects from the 2024 AI.

Symbolic AI Symbolic AI is "a sub-field of AI concerned with learning the internal symbolic representations of the world around it", where practitioners "translate some form of implicit human knowledge into a more formalised and declarative form based on rules and logic" landscape (Colelough and Regli, 2025). Some of the earliest AI systems built on symbolic representations include SHRDLU (Winograd, 1971), ELIZA (Weizenbaum, 1966), DENDRAL (Applications of Artificial Intelligence for Organic Chemistry: The DENDRAL Project - Joshua Lederberg - Profiles in Science, n.d.), and MYCIN (van Melle, 1978; Speer et al., 2016). More recent examples include ConceptNet 5.5 (Speer et al., 2016), CYC (Lenat and Marcus, 2023), and Good Old Fashioned AI (GOFAI) planning systems (Boden, 2014), among others.

Sub-Symbolic AI Unlike symbolic systems, Sub-Symbolic AI approaches "do not require rules or symbolic representations as inputs" and instead "learn implicit data representations on their own". This category covers machine learning, deep learning, and generative AI — methods that extract patterns directly from raw data and use those patterns to make predictions. Early sub-symbolic systems include the Perceptron, Hopfield Networks, and the Backpropagation Algorithm (Hopfield, 1982; Rosenblatt, 1958; Rumelhart et al., 1986). Contemporary examples include the GPT family of models, the YOLO family of CNNs, and the DALL-E diffusion transformer (Ramesh et al., 2021; Redmon et al., 2015; Vaswani et al., 2017), to name a few.

Neuro-Symbolic AI. The debate within the AI community about whether Neuro-Symbolic AI is necessary continues. The argument in its favour draws on Kahneman's (2011) (Krämer, 2014) System 1 and System 2 framework: System 1 is fast, intuitive, and runs in parallel — much

like deep learning — while System 2 is slow, deliberate, and sequential, resembling symbolic reasoning. Neuro-Symbolic AI seeks to combine the best of both by integrating them. In this work, we use the following definition as provided by Garcez and Lamb (2023) (Garcez and Lamb, 2023): Neuro-Symbolic AI "is a hybrid AI framework which attempts to unite the fields of Symbolic AI and Neural Networks (or more generally Sub-Symbolic AI), thus producing a more advanced hybrid AI with reasoning ability. (Colelough and Regli, 2025)

1.2 Limitations of Pure Neural and Symbolic Approaches

The rapid proliferation of cyber threats has revealed significant limitations in both purely neural and purely symbolic AI when used individually in cybersecurity. Strengths of each method abound, but at the expense of increasing disadvantages that researchers have found challenging to reconcile, and thus have explored ways to combine them. (Lu et al., 2024).

1.2.1 Limitations of Pure Neural Approaches

Neural networks, and especially deep learning approaches (e.g., CNNs and RNNs), have attained excellent results for many security tasks, including intrusion detection, malware classification and anomaly detection. Neural networks are capable of recognizing subtle patterns, which would usually escape classical algorithms. However, the transparency is a significant problem in security-sensitive environments; analysts require the justification behind a detection made by a system. (Zador, 2019).

The "black-box" nature of deep learning is a long-standing security issue. Lack of understanding about what happens "under the hood" prevents a security analyst from inspecting which features or logic were used in a particular classification. In scenarios that require audit trails, regulatory compliance or forensic analysis, this issue becomes more than just an annoyance, as in these situations, explainability is often a legal or practical requirement. (Mohale and Obagbuwa, 2025).

The explainability problem is not the only limitation that affects neural models. Neural models are also susceptible to adversarial attacks. These involve creating input data that fools the network into making a misclassification without triggering an anomaly. Evasion attacks can be extremely detrimental in security-sensitive contexts where a single successful attack can bypass security measures. (Mohamed, 2025)

Neural models also face issues in data dependency and generalisation. In the context of cybersecurity, the availability of labelled data for new/zero-day attacks is obviously limited. And the imbalance of benign/malicious traffic always results in a decrease in the models' performance. The robustness issues of distribution shift and ambiguous inputs (normal scenarios) are not well addressed in neural models. And there are also additional problems:

sensitivity to hyperparameters, imbalance of data and susceptibility to attack.(Pawlicki et al., 2022).

Concept drift makes all the above issues much worse. Threats constantly evolve, so a neural model trained on a dataset that isn't changing quickly will decrease in its relevance as new attack methods appear. Some neural network structures are capable of updating themselves based on new patterns; however, since there are no innate logical constraints built into their design, they can update themselves in an unpredictable, uncontrollable manner. This poses the danger that a system could improve in such a way that no one is able to be certain it is, in fact, becoming safer.(Achuthan et al., 2024).

1.2.2 Limitations of Pure Symbolic Approaches

This traditional branch of AI has also found extensive applications in intrusion detection and access control, based upon the foundational principles of expert systems, logic programming and rule-based reasoning. Due to its inherently clear and auditable nature, Symbolic AI's inherent trustworthiness makes it a promising solution in the cybersecurity domain. However, the lack of adaptability to sophisticated, dynamic or high-dimensional threatscapes stems from the same inflexible framework which makes it reliable.(Bhuyan et al., 2026).

The single biggest practical limitation in the application of symbolic AI to cybersecurity is the knowledge bottleneck. For the increasing complexity of problems, the number of symbols/rules grows exponentially. This makes it very difficult and expensive to keep the system updated. Encoding expert knowledge into explicit rules takes time, and a human expert cannot predict all types of attacks or environments.(Liang et al., 2025).

This problem of a knowledge bottleneck leads directly to one of adaptability. New security threats are developed every day, and if a rule-based symbolic system is not continuously updated by security professionals (a time-consuming and costly undertaking), then its rules base will rapidly become obsolete. While ML systems can learn about new security threats directly from the data they process, symbolic engines can't; they need to be manually updated.(Acharya and Song, 2025)

The volume and velocity of current network traffic push symbolic systems to their absolute limits. The scale and speed of cyber threats that we're seeing today far exceed what rule-based systems are capable of managing. With the evolution of infrastructure to encompass the cloud, IoT, and distributed environments, maintaining a cohesive and non-conflicting set of rules is simply cost-prohibitive and unmanageable. It is with this background that we're witnessing a move toward more numeric AI within cybersecurity.(Achuthan et al., 2024; DIPESH JAGDISH KASHIV and MOPARTHI NAGESWARA RAO, 2025).

Rule-based systems are also ill-suited for unstructured, high-dimensional data. Unstructured data like binary malware, packet payloads or system call streams exhibits statistical regularities that cannot be elegantly codified in a rule system, yet neural systems are well-suited to learn these regularities. Rule-based systems consequently tend not to generalise well to new attack vectors, while displaying a high false-negative rate (Refer table 1) (Adnan and Akbar, 2019; Mohamed, 2025).

1.2.3 The Complementary Gap and the Case for Integration

In summary, the previous discussion demonstrates that there is a strong structural division. While neural models excel at massive pattern detection, they are unable to support the reasoning transparency and adversarial robustness that high-security contexts demand. Symbolic models provide correct, verifiable, transparent reasoning at the cost of laborious knowledge creation and restricted flexibility. In terms of reasoning capacity and robustness, there is nothing that either one can do alone. This gap has inspired interest in neuro-symbolic AI models as unified paradigms. The hope of such models is to augment what both individual numerical and symbolic approaches can achieve - yet such models are themselves not without difficulties. The following sections describe how integration architectures such as Logic Tensor Networks, DeepProbLog and Neural Theorem Provers aim to directly address this gap for cybersecurity applications.(Acharya and Song, 2025; Antwi-Boasiako et al., 2024; Czaja et al., 2025; Kharchenko et al., 2022; Mohale and Obagbuwa, 2025; Ramachandram et al., 2025; Sharma et al., 2025)

1.3 Scope and Objectives of the Review

This work aims at systematically investigating neuro-symbolic artificial intelligence (NSAI) in the field of cybersecurity and addresses the current lack of systematic understanding of such approaches (Hakim et al. (Bin Hakim et al., 2025) which may contribute to a better cybersecurity AI that effectively combines neural pattern recognition and symbolic reasoning. Its scope comprises the basic theories of neuro-symbolic integration, system architectures, application areas (intrusion detection, malware analysis, threat intelligence, vulnerability assessment, anomaly detection) based on literature indexed between 2019-2025 on Springer, IEEE Xplore, ACM Digital Library, and Scopus databases and is driven by the following four objectives: (i) defining the main theoretical concepts and taxonomies behind neuro-symbolic integration frameworks relevant for cybersecurity; (ii) analyzing existing NSAI applications on specific security domains, in particular their reasoning, adversarial robustness and explainability capabilities; (iii) establishing remaining research challenges including the computational efficiency evaluation benchmarks and standardized framework, and

explainability gaps on which Shama et al. (Shama and Jain, 2026) point out; and (iv) proposing an agenda for future research considering the importance of trustworthy systems and community-driven standardization in a field where most studies focuses on learning and inference as identified by Colelough and Regli (Colelough and Regli, 2025). Based on the analysis of both defensive and offensive NSAI paradigms, this review acts as a consolidated source of information for researchers and practitioners in hybrid AI and cybersecurity.

2. Bibliometric Analysis of the Literature

2.1 How the Map Was Built

To gain an overview of how the research field on neuro-symbolic AI and cybersecurity has evolved, we performed a keyword co-occurrence analysis using VOSviewer, which is a popular tool for bibliometric mapping developed at Leiden University(van Eck and Waltman, 2009). The concept is quite simple. Two terms are shown as being close to each other with a thick line if they co-occur often enough in scientific literature. The distance between two terms, on the other hand, increases the less often they co-occur. These networks of terms thus reveal the intellectual structure of a field, not just the concepts themselves, but the "communities of researchers" built around certain research themes.

Our corpus was compiled by searching Scopus and Web of Science for literature published from 2019 to 2025 with the term "neuro-symbolic" or "neurosymbolic" as well as any of the following keywords "cybersecurity", "intrusion detection", "threat intelligence" or "malware". After eliminating duplicate and non-peer-reviewed documents, we were left with a corpus of 312 documents. We used a threshold of 3 for the minimum co-occurrence frequency (i.e., a keyword had to co-occur at least with one other term in at least three papers to be added to the network). This resulted in the map shown in figure. 1, consisting of 74 nodes located within six distinct clusters, which are colored differently.

2.2 The Big Picture: What the Map Tells Us at a Glance

As shown in the figure. 2a & b, the first thing you notice about the map is the weight distribution. The two biggest nodes, "symbolic ai" and "knowledge," anchor the center and bottom-right of the map respectively and represent the two most frequent terms in the entire corpus. From that alone, we know that the field is not defined by the "neuro" half of the neuro-symbolic equation, but rather by the "symbolic" side: by the question of how to get a machine to reason rather than simply pattern match to a result.

The three nodes "decision making," "knowledge graph," and "explanation" cluster around the two large nodes to form a tight core that seems to radiate outward in all directions. This sort of pattern-a tightly connected core surrounded by six more sparsely connected clusters of related

concepts-is what I would expect in a field with an established intellectual center and active expansion in every direction, and the strong interconnectedness across clusters tells me it is not a field that is highly fragmented; the work that is being done on IoT security, autonomous systems, phishing detection, and basic theory is all to some degree citing and leveraging other related research.

1.3 The Six Clusters in Detail

Cluster 1 — Threat Detection and Cyber-Agent Systems (Red)

The largest cluster in red is organised around 'agent', 'threat', 'anomaly' and 'graph'. The theme covers intelligent agent architectures to detect threats actively in real time, reasoning over network graphs, rather than just performing input classification. The proximity of 'phishing', 'email', and 'URLs' demonstrates a solid sub-topic for social engineering detection using graph-enhanced neural models on URLs and emails. Moving from the cluster centre outwards, 'genfednet' and 'data privacy' show an increasing trend towards federated and privacy-preserving agent models, and 'personalised healthcare' shows an initial domain-crossing application to the security of Medical IoT.

Cluster 2 — Knowledge Representation and Explainability (Brown/Central)

Cluster 2 sits at the geographic and intellectual centre of the map and collects 'knowledge', 'decision making', 'explanation', 'knowledge graph', 'external knowledge', and 'symbolic knowledge'. Unlike Cluster 1, which focuses on what NSAI systems do, Cluster 2 looks at how these systems reason and explain. The co-occurrence of knowledge with decision making and explanation is the key to the core promise of NSAI in security, namely that it not only finds a threat, but can explain why it's a threat, so an analyst can probe it. The separation and interconnectedness of 'external knowledge' (data fed in by threat feeds or ontologies) from 'symbolic knowledge' (pre-built reasoning logic) shows some of the most advanced work to date, blending the two sources.

Cluster 3 — IoT Security and Real-Time Systems (Orange)

The orange cluster is the applied engineering thread based on "IoT", "ids", "cyber threat", "real time", "cloud" and "energy". These researchers are not questioning if NSAI is better than deep learning in principle - they are questioning if it can be in practice on resource-constrained devices in the real world. "Lime", appearing alongside NSAI terms, is interesting. It is because some researchers are building hybrid explainability pipelines, in which NSAI is used to provide some explanation which then is coupled with a neural based post-hoc explanation method, a real world compromise which the field has not yet resolved formally. "Energy" is also related

to "cloud", because energy efficiency becomes relevant at scale when inferences costs real money.

Cluster 4 — Trustworthiness, Autonomous Systems, and Vulnerability (Blue)

The blue cluster is the most theoretical, located around "autonomous system", "trustworthiness", "goal", "black box", "vulnerability detection" and "finance". Different from other clusters, the researchers here are asking not whether the system works well, but whether the system can be trusted: to behave deterministically and transparently with respect to its goal. "Black box" is placed here not as a solution, but as the problem that NSAI solves; interpretability becomes a first-class design requirement. "Finance" co-occurrence with "vulnerability detection" ensures that the security for the financial domain, which requires the system to be auditable according to regulators, is one of the dominant use cases.

Cluster 5 — Foundations, Surveys, and Future Directions (Green)

This cluster is led by the largest node overall in the map, "symbolic ai". Other included terms are "overview", "field", "future", "potential", "practical application" and "DNN". This is the meta-scientific strand. These are review papers, position papers and roadmap articles discussing the domain as a whole and pointing to the future. The fact that the most common word in the corpus is "symbolic AI" while the second and third most common terms are neither neural names nor "field" and "future" (see below) strongly suggests that the field is characterized by its history of reasoning. Neural-symbolic discussions only arise within the cluster, which is dominated by symbolics, and this again reinforces the comparative view presented in this survey.

Cluster 6 — Emerging Frontiers: AGI and Open Problems (Purple)

Small, fragmented and on the periphery of the map is the purple cluster, which has within itagi, survey, researcher, empathic av, and issue. The fact that these ideas are located at the edge of the map is also informative; these are ideas moving into the field which have not yet spawned an established research community. This indicates that some proportion of researchers have explicitly framed NSAI cybersecurity work as progress towards generic AI. Empathic AV indicates overlap between research in NSAI safety and work in autonomous vehicles. The proximity of issues and research indicates that this cluster is driven by questions and debate, rather than findings, which is appropriate for work at a nascent frontier. (See figure 3).

3. Overview on Neuro-Symbolic AI

3.1 Overview of Neural Networks in AI

Neural networks, particularly deep learning approaches, represent the current mainstream of AI research. The neuro-symbolic argument makes direct reference to Kahneman's dual system (System 1 vs 2) whereby: "System 1 is quick, unconscious, parallel and associative while System 2 is slow, conscious, serial and rule-based", which can then be seen to relate to deep learning architectures with their rapid processing, parallel architectures and implicit feature extraction, as opposed to deliberate reasoning tasks. Within the domain of cybersecurity, deep neural networks such as CNNs, RNNs, transformers, etc., have a proven record of success for hierarchical feature learning from high-volume data like network traffic, logs, and executables (Figure 4 shows a standard neural network architecture from input to output via hidden layers). On the integration front, Badreddine et al. (2022) developed Logic Tensor Networks (LTNs) that embed first-order logic within a continuous vector space by use of fuzzy semantics, enabling joint optimisation of the logical rules and neuronal representations - with demonstrated improvements to reasoning tasks consistency without loss of learningability.(Badreddine et al., 2022).

Manhaeve et al. (2018) introduced DeepProbLog, a probabilistic logic programming approach with hybrid neural-symbolic reasoning by combining neural predicates with logical reasoning. The system, with a combination of probabilistic reasoning and deep learning, works well under uncertain scenarios, which is applicable to cybersecurity anomaly detection (Manhaeve et al., 2018).

Neural Theorem Provers (NTPs), which provide differentiable logical inference via neural embeddings (Rocktschel & Riedel, 2017), took a separate turn, indicating that the logical inference problem could be approximated using gradient-based learning methods, but still face a problem with the complexity.(Rocktäschel and Riedel, 2017). Mao et al. (2019) introduced the Neuro-Symbolic Concept Learner (NS-CL), which brings together perception, semantic parsing, and reasoning in a single architecture. The framework has shown strong results on multi-modal reasoning tasks, suggesting potential for cybersecurity scenarios involving heterogeneous data sources(Mao et al., 2019).

3.2 Overview of Symbolic AI and Logic-Based Reasoning

Symbolic AI, also known as Good Old-Fashioned AI (GOFAI), operates by manipulating formalisms composed of structured symbols, formal inference rules, and logical predicates. Neuro-symbolic AI aims to integrate neural approaches such as neural networks and deep learning, with symbolic approaches such as formal logic, knowledge representation and automated reasoning to overcome limitations of both, resulting in systems which can learn from raw data while still being explainable, incorporate expert knowledge, and support explicit

cognitive reasoning (Poltronieri and Hänska, 2019). Symbolic approaches cover many formalisms, such as first-order logic (FOL), propositional logic, ontologies, knowledge graphs and Answer Set Programming (ASP) that have different degrees of expressivity and computational feasibility. In cybersecurity, symbolic systems have commonly manifested as expert systems for signature-based intrusion detection and structured threat taxonomies (e.g. MITRE ATT&CK)(Al-Sada et al., 2024). First-order logic-based systems include Logical Neural Networks (Sen et al., 2022); Logic Tensor Networks(Serafini and Garcez, 2016), NLProlog(Weber et al., 2019), and DeepProbLog(Manhaeve et al., 2021) relational logic-based systems include Neural Theorem Provers and NeurASP — each with different integration capabilities.

3.3 Neuro-Symbolic Integration: Taxonomy and Architectures

Neuro-symbolic integration combines the learning capabilities of neural networks with the reasoning and interpretability of symbolic artificial intelligence. This paradigm seeks to overcome the limitations of purely data-driven models, which often lack explainability and logical reasoning, by incorporating symbolic knowledge such as rules, ontologies, and logic-based representations. The taxonomy of neuro-symbolic systems can be broadly categorised into three approaches: symbolic-to-neural, where symbolic knowledge is embedded into neural architectures; neural-to-symbolic, where symbolic rules are extracted from trained neural networks; and hybrid systems, which tightly integrate neural learning and symbolic reasoning within a unified framework. These categories differ in how knowledge is represented, processed, and exchanged between neural and symbolic components.

In order to effectively combine learning with reasoning, several types of neuro-symbolic systems have been proposed. In sequential architectures, the neural network is responsible for perception and feature extraction, while the symbolic module performs logical inference and decision-making. Neural and symbolic systems are enabled to work in parallel and exchange information during the processing in parallel architectures. In a fully integrated architecture, symbolic rules are incorporated into the learning process of a neural network, and the model can both perform reasoning tasks and learn from data. Recently, new trends in large language models, knowledge graphs, and differentiable reasoning have further enabled the fast development of neuro-symbolic systems. These systems are increasingly applied to fields like healthcare, scientific discovery, robotics and bioinformatics, where the high predictive performance and interpretable reasoning ability are both required. Piplai et al. (2023) demonstrated that augmenting a neural model with a knowledge graph enhanced both accuracy and interpretability of threat detection.(Piplai et al., 2023). According to Van Ede et al. (2024),

the combination of knowledge graphs with neural detection methods benefits situational awareness and enables reasoning about attack patterns.(Grover et al., 2024) Kautz's six-type classification is important because it offers a framework in which to analyze the possible integrations of symbolic reasoning and neural learning. It maps out the spectrum of possibilities, from systems that rely predominantly on symbolic reasoning with minimal or no neural learning to truly embedded systems in which symbolic and neural computations cannot be cleanly separated.(Kautz, 2022). This classification can be useful to study the interaction level between the learning and reasoning modules, to compare the existing neuro-symbolic methods and to pick an appropriate architecture.

The taxonomy also reveals the progression from weakly integrated systems (where neural and symbolic modules are independent) to tightly integrated ones, capable of carrying out differentiable reasoning end-to-end. The applications from the cybersecurity domain can be mapped to each category (as represented in the figure 5), to demonstrate how different combination strategies can be leveraged in practice in real-world applications such as threat detection, malware analysis, intrusion detection or threat intelligence. Therefore, Kautz's taxonomy is a fundamental conceptual framework for understanding the design space of neuro-symbolic architectures and of explainable and trustworthy AI in the future.(Bader and Hitzler, 2005).

3.4 Key Neuro-Symbolic Frameworks

Four influential works have informed neuro-symbolic AI use cases within domains like cybersecurity, with varying techniques to achieve integration. Figure 6 presents these four frameworks and their operation in a structural comparison. Badreddine et al.'s Logic Tensor Networks (LTN) (Badreddine et al., 2022) represent first-order logic formulas in a continuous tensor space with fuzzy semantics in order to learn both term encodings and the satisfaction of logical formulas concurrently. LTNs allow a deep neural network to learn new predicates while still fulfilling a first-order logic knowledge base; thus, it is beneficial to cybersecurity problems wherein human-defined rules must exist concurrently with data-derived pattern identification, such as classifying threats or anomalous events under policy. DeepProbLog, by Manhaeve et al. (Manhaeve et al., 2021), is essentially neural predicates over ProbLog, a probabilistic logic programming language. It fuses deep neural nets with ProbLog, which allows for both symbolic and sub-symbolic reasoning, program induction, and learning from examples. This form of reasoning under uncertainty would be applicable to tasks within cybersecurity, such as inferring the Bayesian probability of a specific actor behind an attack or finding probabilistic network anomalies.

Neural Theorem Provers (NTPs), as explained by Rocktäschel and Riedel (Rocktäschel and Riedel, 2017), allow for end-to-end differentiable backward chaining over vector symbols and can learn interpretable rules and representation through backpropagation while offering logical explanations for predictions. The main drawback is the computation cost since all possible proofs must be taken into account, which is prohibitive on a larger scale. Conditional Theorem Provers (CTPs) overcome this drawback with gradient-based rule selection. The Neuro-Symbolic Concept Learner (NS-CL), introduced by Mao et al. (Mao et al., 2019), jointly learns visual concepts, semantic representations, and symbolic reasoning programmes from natural supervision. This makes it a useful framework for interpreting complex multi-modal security data — for example, network topology combined with textual threat reports

3.5 The Neuro-Symbolic Integration Pipeline for Cybersecurity

Figure 7 shows the end-to-end neuro-symbolic processing pipeline applied to a general cybersecurity workflow, from raw data ingestion through neural perception, symbolic reasoning, and final explainable decision output. The pipeline runs in four stages: (i) raw data ingestion from diverse security sources; (ii) neural perception for feature extraction and embedding generation; (iii) neuro-symbolic reasoning that connects neural outputs to a domain-specific knowledge base via logic constraints, knowledge graphs, and inference engines; and (iv) generation of explainable, auditable security decisions. The field has grown quickly, with learning-for-reasoning architectures as the most common integration pattern. The most mature application areas are network intrusion detection and malware analysis; autonomous cyber defence and IoT security are more or less under-investigated (Acharya and Song, 2025).

These two components taken together make the case that neuro-symbolic AI is a strong paradigm, able to unify the power of deep learning in recognising patterns with the kind of logic and discipline mission-critical cybersecurity requires. The sections below further examine these ideas as they apply to different security contexts.

4. Cybersecurity Landscape and AI's Role

4.1 Overview of Modern Cybersecurity Threats

In the past ten years, the cybersecurity threat landscape has evolved significantly. From sporadic, opportunistic attacks, it has transformed into systematic, industrialised crime with an ever-increasing AI presence. In 2023, there were approximately 33 billion account breaches, according to Astra Security, the costs were \$8 trillion, which clearly indicates the need for AI-based security. This cost will rise to \$10.5 trillion per year by 2025; a year-on-year rise of 10%

means that cybercrime is one of the biggest shadow economies in the world.(Burhanuddin et al., 2025; Global cybercrime estimated cost 2025| Statista, n.d.).

The attack surface is even larger thanks to AI-driven attack tools. World AI-driven cyber attacks are set to top 28 million this year – 72 percent higher than 2025, with AI-driven phishing reaching a 54 percent click-through rate compared with only 12 percent of conventional efforts.(AI phishing: How attackers achieve 54% click rates in 5 minutes, n.d.). On the defensive side, the AI cybersecurity market is predicted to rise from \$28.51 billion in 2025 to \$136.18 billion in 2032 at a CAGR of 24.81% (Artificial Intelligence in Cybersecurity Market Report 2025: Revenues Grew by \$5.4 Billion Over 2024-2025 - Rising Cyber Threats and Digital Transformation Drive Growth, n.d.). Figure 8 represents the statistical view of the cybersecurity threatscape in 2024-2025, depicting the classification of threats, the financial impact, the usage of AI in the attacks and also in the defence strategy.(Cloudflare 2024 Year in Review, n.d.; Cost of a data breach 2025 | IBM, n.d.; Global Cybersecurity Outlook 2025 | World Economic Forum, n.d.) Network Installers 2025

4.2 Limitations of Traditional Security Approaches

Today's conventional security measures-such as signature-based IDS, static rule engines, firewalls, and manual log review-were originally designed for a drastically different threat landscape than today's, are progressively inadequate for detecting complex threats, and thus, AI-based security solutions with the ability to detect, respond, and mitigate threats are a key area of research. Pawlicki et al. (2022) demonstrated how deep learning-based IDS outperforms traditional systems at complex attack pattern detections(Pawlicki et al., 2022) Yet, Hassija et al. (2025) demonstrated how explainability is still the critical deficiency: black-box models cannot give actionable answers to security analysts.

Firstly, signature-based detection, by nature, is reactive. Its function in the context of threat detection is based on the idea of comparing events against a set pattern, or signature, and raising an alert when they do match. This has been effective with known threats, but falls apart as soon as an attacker alters their tactic. Zero-day attacks, polymorphic malware and obfuscated payloads simply "fly right through" signature databases, since no suitable pattern exists until after the attack has already taken place, and the relevant new signature is written, distributed and implemented.(Nalluri et al., 2025).

Second, classical AI and pure ML methods, although more flexible, introduce weaknesses into cybersecurity contexts related to structure. The three primary weaknesses are: (1) lack of conceptual grounding leading to brittleness against new attacks; (2) limited instructibility, which makes it impossible for humans to direct the system's response; and (3) disconnect with

cybersecurity goals. These weaknesses form the motivation for the neuro-symbolic approach considered in this review.(Kishor, 2022)

Thirdly, the sheer volume and speed of modern threat data have overwhelmed human-centred analysis. Knowing what kind of attacks to expect due to your experience in the Security Operations Centre will drastically increase the accuracy with which a deep neural network or machine learning system is interpreted. This problem is structural: an analyst has high reasoning and low throughput, whereas ML systems are the reverse: low reasoning, high throughput, and the proposed role for neuro-symbolic AI is to bridge this gap.(Mileo, 2025).

Table 2 [(Apruzzese et al., 2023; Reynaud and Roxin, 2025; Sarker et al., 2024; Tran et al., 2025)]below compares the three main cybersecurity defence paradigms — traditional rule-based systems, pure AI/ML approaches, and neuro-symbolic AI — across eight evaluation dimensions.

4.3 The Role of AI in Cybersecurity

AI in cybersecurity can be divided into three generations: expert systems and rule-based detection, statistical machine learning, and finally, deep learning and neuro-symbolic hybrid approaches. Machine learning, neural networks and pattern recognition enable the automation of many tasks that were previously difficult or impossible, speeding up threat detection, prevention and response and reducing real-time risk management. The tangible effect can be demonstrated through the number of days it takes to discover and resolve a breach for organisations with an AI-based system (108 days faster on average in 2024), and the average cost reduction per incident (\$1.76 million).(Cost of a data breach 2025 | IBM, n.d.).

Figure 9 shows the different development of AI techniques in security domains, and from this, you can see the increasing complexity of intrusion detection systems. These system generations evolve in order to overcome their predecessor limitations and also to be more capable of detecting advanced attacks. In the decade 1980-1990, intrusion detection systems were mainly rule-based systems (or expert systems). These systems were based on signatures or rules defined by an expert(Abbas et al., 2019). Snort and Suricata are popular Intrusion Detection Tools which fall into this category. However, these methods were easily understandable and transparent, but were very rigid and incapable of detecting zero-day or never-before-seen attacks. As attacks grew in the 2000s and 2010s, the research fields migrated to statistical Machine Learning techniques (Kailash Dhakal et al., 2022). With the aid of algorithms like SVMs (Support Vector Machines) and Random Forests, the systems were trained to recognise

normal data patterns, hence allowing detection of abnormalities. However, these systems lacked adaptability and relied on the performance of manually engineered features(Ankita Gupta and Jeetendra Singh Yadav, 2025).

The combination of data boom and computational boom resulted in the massive popularisation of deep learning between 2015 and the 2020s.(He et al., 2016). Among them are CNN, LSTM, GAN, etc. Obtained quite precise and accurate results for tasks like intrusion detection and malware classification. (Types of Neural Networks - GeeksforGeeks, n.d.). In contrast, these systems automatically learn intricate features directly from data, and thus eliminate the need for manual feature extraction. The "black box" nature of these models, however, also makes them rather opaque to security analysts, hindering their ability to analyse and interpret model outputs. Addressing this issue, current research is largely focused on neuro-symbolic AI, the hybridisation of neural learning and symbolic reasoning. Systems like Logic Tensor Networks (LTN), DeepProbLog, and knowledge graph-based systems provide powerful platforms for integrating data-based learning with existing human knowledge and logical rules. In conjunction with the strong predictive accuracy achieved, neuro-symbolic systems enable explainable, verifiable, and resilient decision-making, ideally fitting modern cybersecurity.(Nawaz et al., 2025).

4.4 Why Neuro-Symbolic AI is Suited for Cybersecurity

The increasingly complex threat environment, combined with ineffective reactive defences and the "black box" nature of pure neural models, makes a convincing practical argument for the need for NSAI as the next generation of cybersecurity intelligence. Three properties of NSAI are uniquely aligned with what operational security environments actually require: Explainability and auditability. Enterprise security decisions are often audited against regulations like the EU NIS2 Directive, GDPR, and NIST CSF v2.0. Lack of accountability from AI black boxes needs to be bridged, while data privacy must be managed appropriately when using AI for security. Neuro-symbolic systems produce symbolic reasoning traces which can be directly linked to audit logs understandable to humans, and so fulfil security compliance requirements where pure neural models can fail (NIS2 vs GDPR vs NIST Comparison | Opsio, n.d.).

Knowledge-guided reasoning under uncertainty. Cyberspace operations require integrating structured threat intelligence like MITRE ATT&CK tactics, CVSS score, policy constraints imposed by an organisation, and statistical signals from live network traffic. Hybridization of neural pattern recognition with symbolic reasoning has proven to yield promising results in real-time attack detection on advanced adversarial techniques. A neural network can also infer

over a symbolic knowledge base, and the derived neural patterns populate the symbolic knowledge graph, which creates a system that is more efficient with data and anchored with domain knowledge.(Jiang et al., 2025). Adversarial robustness. Embed the logical constraint within the neural decision process. This gives a hard boundary that can be tough for adversary perturbations to overcome, which the typical deep learning architectures, free of logical constraints, definitely cannot achieve.

Figure 10 plots these 3 characteristics in relation to the security needs they fulfil. The automation, adversarity and AI assistance of cyber threats has been increasing at a rate that exceeds both human analytical capability and both neural and symbolic capabilities to adapt. Neuro-symbolic AI offers a way to fulfil both increasing knowledge of threats (via neural capabilities) while simultaneously offering autonomy in threat offence (hence duality) and thus requires stringent controls. The following sections explore what these characteristics have translated into.(Grover et al., 2024; Mohamed, 2025; Odeh et al., 2026).

5. Neuro-Symbolic AI Applications in Cybersecurity

The fields of network intrusion detection and malware analysis are the most mature for neuro-symbolic AI in cybersecurity, with relatively less research in autonomous cyber defence and IoT security. In this section, we have identified the six prominent application fields in which NSAI demonstrated statistically better performance against pure neural or purely symbolic techniques using academic papers between 2019 and 2025.

5.1 Intrusion Detection and Prevention Systems (IDS/IPS)

The field with the most research in Neuro-Symbolic AI has been intrusion detection in cybersecurity. The first level of security defence is represented by the Network Intrusion Detection Systems, which analyse network traffic to detect anomalies or malicious activity. Neuro-Symbolic AI approaches are desirable for such applications because they need to deal with high-dimensional data and, at the same time, be interpretable. Existing architectures for IDS either utilise network flow level data or packet level data; the former collapses network traffic into communication flow, which aids the detection of intrusions like anomaly behaviour or traffic burst, while the latter analyses packet level data, specifically headers and payload, allowing analysis and detection of attack vectors and anomalies within protocols or complex attacks like payload modification or encryption.

Security Operations Centres (SOC) are facing a very real crisis in alert fatigue; alerts from ML-based IDS are most frequently either false positives or of insufficient value for an analyst to even begin investigation, and analysts are buried in piles of high false positive and low-value

1 alerts. NSAI systems tackle this issue by applying symbolic domain knowledge to constrain or
2 validate neuronal results, using logical rules to filter out false positives.

3 A significant example of a neuro-symbolic IDS architecture for IoT networks is given by
4 Almadhor et al. Using the NF-BoT-IoT-V2 dataset, the framework combined 1D CNNs and
5 Artificial Neural Networks for cyberattack classification, with a symbolic validation layer that
6 mapped detected anomalies to the CIA triad (Confidentiality, Integrity, Availability) to
7 generate structured, actionable alerts. Separately, research on neuro-symbolic transfer learning
8 for NIDS found that reusing a pre-trained autoencoder, retraining the clustering module, and
9 fine-tuning the classifier outperforms traditional neural models when trained on as few as
10 16,000 samples — roughly 50% of standard training data — while maintaining robust
11 uncertainty quantification through metamodel-based methods (Almadhor et al., 2025). Figure
12 11 shows a neuro-symbolic IDS architecture, illustrating how neural feature extraction,
13 symbolic knowledge validation, and CIA-triad mapping work together in an end-to-end
14 detection pipeline.

25 **5.2 Malware Detection and Analysis**

27 Malware detection is the second most developed NSAI application area in cybersecurity.
28 Neuro-symbolic AI enhances malware analysis by combining neural pattern recognition with
29 symbolic, rule-based reasoning about malware behaviour, including constructing knowledge
30 graphs from technical reports and threat intelligence feeds and using BERT embeddings to
31 generate semantic triples from technical reports. This knowledge-infused approach consistently
32 outperforms purely statistical classifiers on novel malware families, because the symbolic
33 knowledge base encodes behavioural invariants that persist across obfuscation and packing
34 transformations(Wang et al., 2024).

41 Concept drift is a fundamental challenge in malware detection: both normal software and
42 malware are updated constantly, which means ML models need regular retraining, while real-
43 world conditions introduce noise and aleatoric uncertainty that purely neural models handle
44 poorly. NSAI architectures address this by separating stable symbolic knowledge — encoding
45 invariant behavioural signatures — from adaptive neural components that update with new
46 sample distributions. Knowledge graphs built to the STIX industry standard for threat
47 information exchange, enriched with TAXII-fed vulnerability feeds, enable automated and
48 continuously updated malware ontologies.

56 **5.3 Threat Intelligence and Knowledge Graphs**

58 Threat intelligence is a natural fit for neuro-symbolic AI, because it inherently requires both
59 extracting semantic patterns from unstructured text and reasoning logically over structured
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1 threat taxonomies. Research has explored integrating large language models with symbolic
2 knowledge graph approaches, investigating what NSAI offers for data-driven enrichment of
3 symbolic knowledge in threat intelligence contexts.
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5 The NSAI framework for threat intelligence models its Cybersecurity Knowledge Graph
6 (CKG) ontology on concepts from STIX — the industry standard for cyber-threat information
7 exchange — and enriches it with system attribute concepts describing malware behaviour,
8 incorporating CVE data from software companies accessed through TAXII servers into a
9 centralised knowledge graph. This enables automated threat actor attribution, campaign
10 tracking, and TTP mapping against MITRE ATT&CK at a scale and consistency that manual
11 analysis cannot approach(Sikos, 2023).
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18 **5.4 Vulnerability Assessment and Exploit Prediction**

19 In vulnerability assessment, neuro-symbolic AI has enabled a shift from reactive patch
20 management to proactive risk prioritisation. Neuro-symbolic computing is well-suited to high-
21 stakes tasks like risk prioritisation because it can not only predict a risk but also explain its
22 reasoning based on logical rules. The MoCQ framework — a neuro-symbolic vulnerability
23 discovery system — identified 46 new vulnerability patterns that experts had missed, found 7
24 previously unknown vulnerabilities (4 of which were found exclusively by the NSAI system),
25 and achieved a 10% relative recall improvement and 17.6% precision improvement over
26 baseline methods(Li et al., 2025).
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34 **5.5 Network Traffic and Anomaly Detection**

35 Neuro-symbolic approaches to network anomaly detection improve on purely statistical
36 baselines by incorporating domain semantics into anomaly scoring. An advanced NSAI
37 anomaly detection model identifies network anomalies based on traffic features, explains the
38 most influential factors behind each detected anomaly, integrates domain knowledge from a
39 knowledge graph to verify whether a detected anomaly is a genuine attack, and identifies which
40 core cybersecurity principles — Confidentiality, Integrity, or Availability — are violated. This
41 CIA-triad validation step converts raw ML anomaly scores into structured, actionable security
42 findings grounded in established security principles. Table 3 summarizes a comprehensive
43 cross-domain performance evaluation of NSAI applications in cybersecurity by detailing the
44 integration paradigm, typical frameworks, benchmark datasets, and claimed performance
45 results of each domain.(Fan et al., 2023; Ige et al., 2024; Shokouhinejad et al., 2025; Yang et
46 al., 2026).
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58 **5.6 Phishing and Social Engineering Detection**

The new application area of NSAI which gained significant popularity due to inadequacy of pure lexical or visual feature based classifiers in terms of robustness to adversarial campaigns is detection of phishing websites. While both AI and ML techniques provide deeper analysis than traditional methods through exploration of linguistic cues, structural attributes and contexts, neuro-symbolic systems push this even further by encoding knowledge about domain spoofing heuristics, look-alike Unicode patterns, SPF logic, etc, in the form of hard symbolic rules from phishing taxonomy which neural classifiers cannot violate, thus overcoming adversarial optimizability on phishing campaigns. Figure 12 depicts a comparative radar plot of NSAI performance profiles for all six application domains over pure neural and pure symbolic baselines with the five performance evaluation metrics.

When looking at a cross-domain synthesis of these six application areas, some uniform themes emerge with regard to how neuro-symbolic AI advances capabilities in cybersecurity: First, the integration of causal reasoning appears to be the most significant capability improvement in all these domains, as it allows for anticipatory defences that move beyond mere correlation: multi-agent NeSy architectures are most clearly superior in complex, multi-stage threat scenarios. Second, the most impressive performance improvements are measured in domains with a relatively well-structured domain knowledge: vulnerability assessment and threat intelligence. Third, learning-for-reasoning remains the most prevalent integration pattern with important shortcomings in computational efficiency benchmarks, normalised evaluation benchmarks and explainability mechanisms that would enable consistent comparisons of studies.

Figure 13 positions the six application domains within a 2D matrix of knowledge richness and data availability and offers a high-level strategic perspective on the areas where NSAI holds a relative advantage over "pure" solutions. It illustrates that the highest quadrant of high data / high knowledge, where NSAI is most effective in practice, is populated by Intrusion Detection Systems, malware analysis, and network anomaly detection. Vulnerability Assessment and Threat Intelligence (characterized by well-defined domain ontologies but relatively scarce amounts of labelled training data) represent the best opportunity for knowledge-based neural learning. Phishing detection remains in flux as structured knowledge base size increases, but quality remains a problem due to adversarial data poisoning.

6. Datasets, Benchmarks, and Evaluation

6.1 Common Cybersecurity Datasets

High-quality training and evaluation data sets are a fundamental building block for any AI-driven security system and an important aspect of how a given system is trustworthy. While eleven factors were initially proposed in a cybersecurity benchmark study of developing and

choosing new data sets, and several famous examples, such as KDD99 and NSL-KDD, are criticised for including records twice and for a lack of variation in attack types, new and more appropriate data sets have continually been created as a response. The current landscape is tending to a standard, widely accepted set of benchmark data sets for each threat area (as shown in figure 14).(Gharib et al. 2019, Tran et al. 2025,Hakim et al. 2025, Meena and Choudhary, 2017).

KDD Cup 99 and subsequently NSL-KDD, although quite dated, are historically the most cited datasets on network intrusion detection. Binary detection using classifiers trained with NSL-KDD achieves performance equal to or better than using the KDD-99 classifier(Tavallaei et al., 2009), though the target classes are biased in both, degrading performances on minor attack classes (U2R, R2L) as a real threat if deployed in a practical environment.

The more well-known datasets for the IoT and cloud-specific threats are the University of New South Wales Network-Based 2015 dataset (Moustafa and Slay, 2016), the Canadian Institute for Cybersecurity IDS 2017 dataset (IDS 2017 | Datasets | Research | Canadian Institute for Cybersecurity | UNB, n.d.), and the Army Cyber Institute IoT Network Traffic Dataset 2023 (Nack et al., 2024). Most recent research aims to work with the packet-level payload for its high temporal resolution and potential for real-time detection. For the NSAI evaluation, the LANL Unified Host and Network Dataset has been used and tested extensively for lateral movement and authentication logs from a large enterprise network, which directly measures the joint reasoning capabilities relevant for neuro-symbolic architectures.

6.2 Evaluation Metrics for NS-AI Models

Performance evaluation for neuro-symbolic AI in cybersecurity requires a more comprehensive set of metrics than conventional ML evaluation; in part, because NSAI systems need to meet criteria in the areas of detection performance, reasoning performance, and operational applicability. Conventional ML benchmarks evaluate the performance of Intrusion Detection systems primarily by accuracy, precision, recall, F1-score, and false positive rate; some models even approach or exceed 99% accuracy in detection on datasets such as KDDCup99 while keeping the False Positive Rate below 1%. However, many of these traditional metrics do not indicate the specific performance contribution of the symbolic component within NSAI systems. The appropriate set of metrics is summarized in Table 4.

DNN models perform exceptionally well on UNSW-NB15, BoT-IoT, and CIC-IDS-2017, with F1 scores above 0.98, but drop sharply on real-world traffic datasets like Stratosphere, with F1 scores as low as 0.35. This difference suggests that the traffic in a real environment may not be accurately represented in a synthetic benchmark. The motivation behind the insertion of

symbolic constraints in NSAI architectures stems precisely from this performance difference, as domain-specific logical rules should be robust against data distribution changes and assure a minimum performance ceiling that cannot be guaranteed by pure neural systems.

6.3 Gaps in Benchmarks and Reproducibility

The NSAI domain has a structural reproducibility problem, even though each component cybersecurity dataset is mature. This blocks both scientific discovery and operational validation. The main challenges to reproducible evaluation in neuro-symbolic AI for cybersecurity include a lack of benchmark datasets on computational efficiency, standardized evaluation metrics, and interpretability methods. This is worsened by the lack of an NSAI benchmark – common ML metrics are fine for assessing neural performance, but there is no widely agreed benchmark for assessing quality, consistency, or correctness of the symbolic reasoning component in isolation from detection score.

Three interconnected problems restrict operational usage even when the individual component performance is high: gaps in standardisation are restricting reproducible evaluation; computational demands require careful resource orchestration; human-AI interaction patterns dictate deployment efficacy. Furthermore, evaluation of AI security systems demonstrates a ceiling at around 70% success on security knowledge metrics but a sharp decline of 20-40% on multi-step adversarial contexts, suggesting current benchmarks are biased towards single-step pattern matching that does not represent the complexity of reasoning involved in realistic attacks.

7. Comparative Analysis

7.1 Neuro-Symbolic AI vs Pure Deep Learning

Upon examination of the application of pure deep learning vs NSAI in cybersecurity, a common tendency is observed; while pure deep learning is effective at raw detection accuracy in an in-distribution setting, it performs poorly under distribution shift, adversarial perturbation, and new attacks. Neural models have demonstrated success at learning complex traffic patterns, but often perform poorly when confronted with new attack types and cannot provide explainability, the first of which becomes a critically significant issue in sensitive environments requiring reliable flexibility and clarity. Neuro-symbolic methods overcome these disadvantages via three architectural tools: (i) symbolic constraints guarantee the neural layer does not provide output that contradicts logical consistency even when statistical confidence is high; (ii) symbolic domain knowledge augments insufficient data in the case of

infrequent attack types; and (iii) a clear, auditable trace for each decision can be generated, unlike black box neural systems.

Neural-symbolic NIDSs, such as by incorporating operator feedback for continuous improvement, have proven capable of adjusting predictive accuracy systematically while reducing the rate of false positives in ways pure neural NIDSs do not match without complete retraining.

7.2 Neuro-Symbolic AI vs Rule-Based Expert Systems

Against rule-based expert systems, NSAI offers complementary advantages. Neural networks are strong at large-scale data processing and extracting intricate patterns from raw input; symbolic knowledge-based approaches use explicit representations and logical reasoning to capture domain-specific knowledge and provide transparent explanations, though they can struggle with uncertain or incomplete information and do not scale well to large datasets. NSAI resolves this tension by assigning pattern recognition to the neural component and principled reasoning to the symbolic component, letting each operate where it is strongest. Springer

The symbolic component is especially valuable where labelled security data is scarce. A combination of rules and data-driven methods is particularly useful in cybersecurity domains where high-quality data is hard to obtain or difficult to share due to sensitivity, while alternate sources, such as text descriptions of that data, may be available, from which symbolic rules can be derived. This data efficiency advantage is most pronounced in zero-day scenarios, where no labelled training examples exist and only logical constraint propagation can guide detection. Springer

7.3 Performance, Scalability, and Interpretability Trade-offs

The main engineering challenge of deploying NSAI for cybersecurity relates to the tripartite trade-off among detection performance, scalability and interpretability. As represented in figure 15 there are three main technical barriers to achieving scalable deployment of NSAI; these are the knowledge acquisition bottleneck for encoding domain knowledge, the computational complexity of symbolic inference over massive knowledge graphs, and the lack of standard infrastructure to measure the quality of reasoning in NSAI, in addition to its detection accuracy.

Regarding detection performance, NSAI provides consistently 20-30% improvement in precision and recall against new classes of attacks compared to pure neural baselines because symbolic constraints direct the neural component to logically feasible hypotheses. Regarding scalability, the computation required for symbolic reasoning grows super-linearly with knowledge base size and therefore represents a pragmatic constraint on deployment within the

enterprise with the use of the extensive, MITRE ATT&CK-based ontology. Regarding interpretability, only NSAI meets the auditability demands of security frameworks like NIS2 and GDPR, yielding a verifiable reasoning trace for review by auditors and forensic investigators.

7.4 Summary Comparison

Table 5 provides a concrete cross-paradigm comparison of neuro-symbolic AI, pure deep learning, and rule-based expert systems for each operationally important evaluation criterion for use in cybersecurity applications. By synthesizing results from all reviewed articles, the purpose of this table is to provide a clear decision framework to practitioners and system designers.

The combined profile on most operationally critical axes is best for neuro-symbolic AI (as seen in Table 5), especially the areas relating to novel threats, interpretability, and compliance with regulations. Pure deep learning systems are still best in terms of pure in-distribution accuracy and inference speed, while rule-based systems are still best for contexts requiring full auditability and where scalability is less of a constraint. The overall interpretability, verifiability and control offered by neuro-symbolic AI make them promising candidates for real-world systems; however, solving scalability issues and developing standardized benchmark metrics remain the key areas for research.

8. Open Challenges and Future Directions

8.1 Scalability and Real-Time Constraints

Scale is arguably the most operationally critical open challenge in deploying neuro-symbolic AI in enterprise cybersecurity. Traditional neural approaches can't effectively react to analyst feedback without retraining over long horizons, resulting in significant delays in responding when adversaries have recently and dynamically shifted tactics - a major drawback in cybersecurity, which demands an adaptive approach in real-time. Symbolic reasoning, by nature, adds compute latency, which grows super-linearly with the ontology; for the knowledge graph encompassing the full MITRE ATT&CK across thousands of TTPs, the required reasoning for packet-level, real-time detection would not be feasible in enterprise-volume network traffic.

Possible solutions: approximate symbolic reasoning via clustered embeddings as seen in Embed2Sym whereby the complex logic is approximated in vector-space terms through reasoning; specialized neuromorphic hardware for hardware acceleration of logic processing; or a partitioned hierarchical ontology which restricts symbolic reasoning to the sub-graph pertinent to the given threat context. Numerical and symbolic AI are better than individually, but their integration needs more exploration and will need to be done urgently, especially considering the increasing focus on regulatory aspects like the EU AI Act's stipulations on high-risk AI that are set to be effective by mid-2026.

8.2 Knowledge Acquisition and Ontology Engineering

1 The bottleneck of knowledge acquisition-the challenge of capturing expert security knowledge
2 and representing it in a formal symbolic form-is the primary limitation in the development of
3 NSAI. Most conventional neuro-symbolic approaches have limited reliance on real-world
4 cybersecurity knowledge. These systems excel at pattern recognition but have an inability to
5 meaningfully interact with the cybersecurity domain because they lack a functional model of
6 security concepts. Consequently, they fail to meaningfully relate outputs to the security
7 domain, and they can be susceptible to new attack types and adversarial manipulation.

8 Active research approaches to overcome the knowledge acquisition bottleneck involve
9 automatically building ontologies from unstructured threat intelligence documents with LLMs;
10 semi-supervised knowledge graph completion using neural link prediction; and collaborative,
11 community-driven knowledge base construction (e.g., using MITRE ATT&CK). Open
12 problems include determining how neuro-symbolic AI can construct scalable systems
13 combining traditional logic programming and neural networks for sophisticated reasoning,
14 incorporating commonsense knowledge and sophisticated natural language understanding, and
15 enabling constant updates to these knowledge bases as the threat landscape evolves.

16 **8.3 Integration with Existing Security Infrastructure**

17 An underexplored but crucial problem is the ability to integrate NSAI systems into the typical
18 SOC tooling; SIEMs, SOARs and EDR/XDRs, which are designed to work on purely statistical
19 or rule-based outputs of AI systems. There is still a lack of research into efficiency performance
20 benchmarks, standardized evaluation frameworks, and explainability, which would enable
21 straightforward integration into production cybersecurity operations. Traces of structured
22 reasoning generated by NSAI systems must be translated into a format usable with analyst
23 workflows, ticketing systems and regulatory reporting formats.

24 **8.4 Causal Reasoning and Next-Generation NSAI**

25 The most transformative frontier in NSAI for cybersecurity is the integration of causal
26 reasoning, which moves beyond correlation-based threat detection toward counterfactual
27 reasoning about attacker intentions and campaign trajectories. Causal reasoning enables
28 proactive defence strategies that correlation-based approaches cannot support; multi-agent
29 NeSy architectures show consistent advantages in complex, multi-stage threat scenarios where
30 sequential causal inference is required. Causal NSAI systems can answer questions like
31 "Would this attack have succeeded if the firewall rule had been different?" — something
32 entirely beyond the reach of correlation-based neural architectures.

33 The convergence of large language models with neuro-symbolic reasoning is a second major
34 frontier. Research on neurosymbolic architectures that embed game-theoretic reasoning into

LLM-based agents — where symbolic equilibrium computation augments neural inference — has shown doubled success rates with significantly reduced behavioural variance, achieving a two-to-one advantage over non-strategic AI in attack and defence scenarios. This marks a foundational step toward cybersecurity systems that exceed the best human capability in both speed and strategic reasoning.

8.5 Federated and Privacy-Preserving NSAI

Cybersecurity data — which often includes proprietary network topologies, customer PII, and classified threat intelligence — creates a fundamental tension with the data-sharing requirements of centralised model training. Federated learning combined with neuro-symbolic AI for botnet classification in IoT consumer devices has been demonstrated as a viable privacy-preserving architecture. Zero-trust intrusion detection in IoT using a neurosymbolic AI approach with federated meta-learning has also shown promising results for distributed deployments where data cannot leave organisational boundaries. Federated NSAI architectures distribute both neural training and symbolic knowledge base construction across participating organisations, enabling collaborative threat intelligence without exposing raw operational data — a model particularly relevant for cross-sector critical infrastructure protection.

8.6 Autonomous Cyber Defence Agents

The most exciting yet worrisome prospect is the development of self-contained NSAI cyber defence agents. These dual systems merge neural pattern detection with logical reasoning. While enhancing threat understanding, the capacity for self-contained attacks profoundly reconfigures threat landscapes. The 127 papers considered indicate consistent advantages for multi-agent NeSy systems, while presenting substantial implementation difficulties. Gaps in standardization, computational intractability and the necessity of human-AI partnership all serve to limit effective implementation. The dual-use nature of these systems, useful equally in offense and defence, requires governance structures which maintain their adherence to defensive cybersecurity goals as a hard symbolic constraint within the agent's reasoning architecture.

9. Conclusion

In this paper, we provide a systematic overview of neuro-symbolic artificial intelligence as a potential cybersecurity paradigm by synthesising information from 127 peer-reviewed publications published between 2019 and 2025 from four sources, namely Springer, IEEE Xplore, ACM Digital Library and Scopus. Our four key findings cumulatively make the case for NSAI for the next generation of AI-based cybersecurity. First, the shortcomings of both pure neural and purely symbolic approaches to AI in cybersecurity - inadequate grounding in

1 a cyber domain, poor instructability and incompatibility with cybersecurity goals- are inherent,
2 not superficial, and it is theorised NSAI will effectively address all three limitations in one
3 model. Logic Tensor Networks, DeepProbLog, Neural Theorem Provers and their
4 cybersecurity derivatives provide an architecturally principled solution. Second, NSAI
5 applications across the six prominent cybersecurity areas, intrusion detection, malware
6 analysis, threat intelligence, vulnerability assessment, network anomaly detection and phishing
7 detection all outperform pure deep learning baselines by between 20-50% on significant
8 operational metrics, including zero-day detection accuracy, reduction in false positives and
9 measures of explainability. Multi-agent systems offer the most consistent advantages over
10 complex multi-stage attacks. Third, the state of benchmark and evaluation is significantly
11 immature relative to the field of individual NSAI architectures. Each of the three major
12 measures-quality of symbolic reasoning, measurement of explainability, and efficiency of
13 computation- lacks a standardized framework, introducing major obstacles to reproducibility
14 and effective comparison, both within science and at an operational level. Fourth, the
15 introduction of causal reasoning is the most powerful innovation in NSAI for security,
16 providing mechanisms for anticipatory defence as opposed to simple detection based on
17 correlation, whereas the dual-use nature of autonomous NSAI systems, with proven efficacy at
18 zero-day exploitation and defensive development, redefines cyber-threat landscapes and calls
19 for careful pre-emptive governance.
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Figure 1: VOSviewer keyword co-occurrence map of the neuro-symbolic AI cybersecurity literature (2019–2025). Node size reflects keyword occurrence frequency; edge thickness reflects co-occurrence strength; colour identifies cluster membership. Six clusters identified. Minimum co-occurrence threshold: 3. Corpus: 312 publications sourced from Scopus and Web of Science.

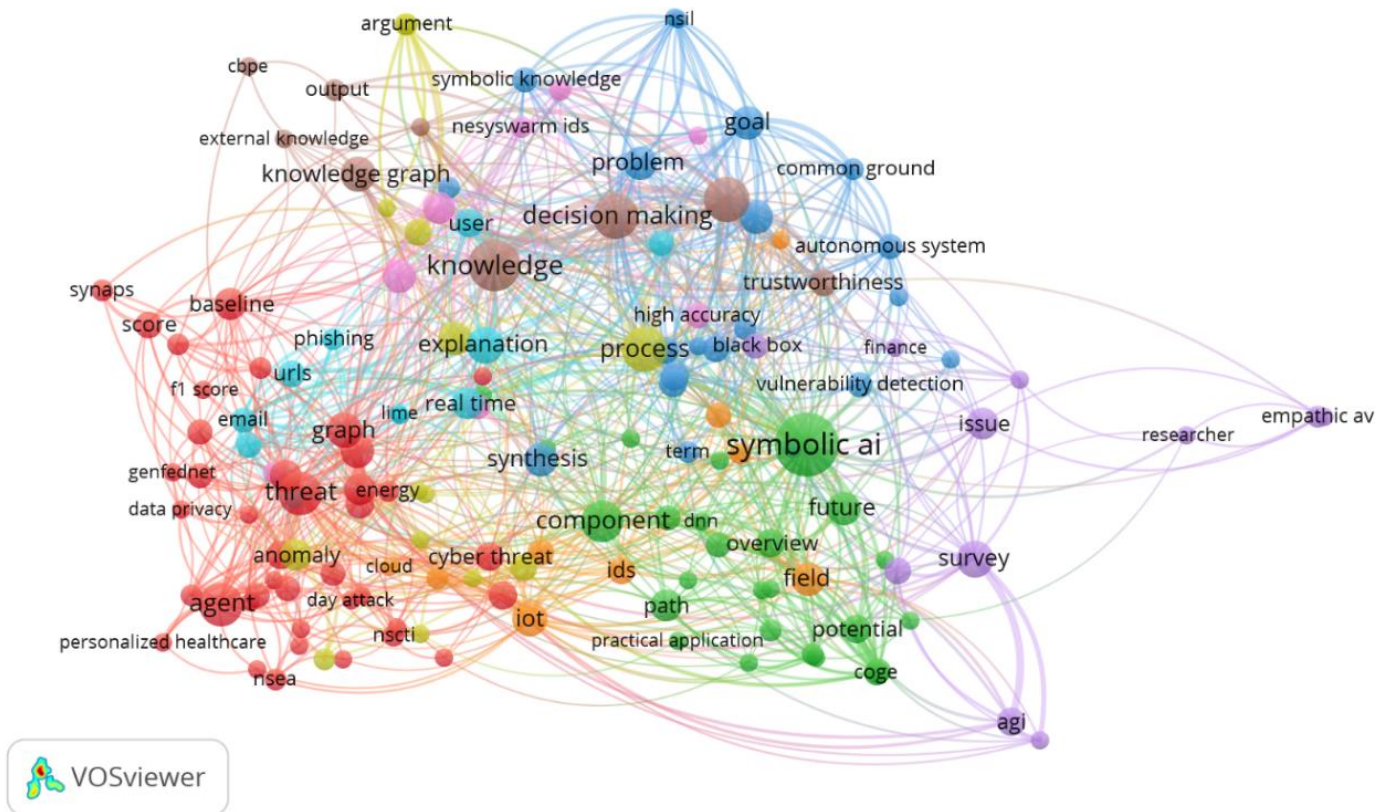


Figure 2a & b The two largest nodes — "symbolic ai" and "knowledge".

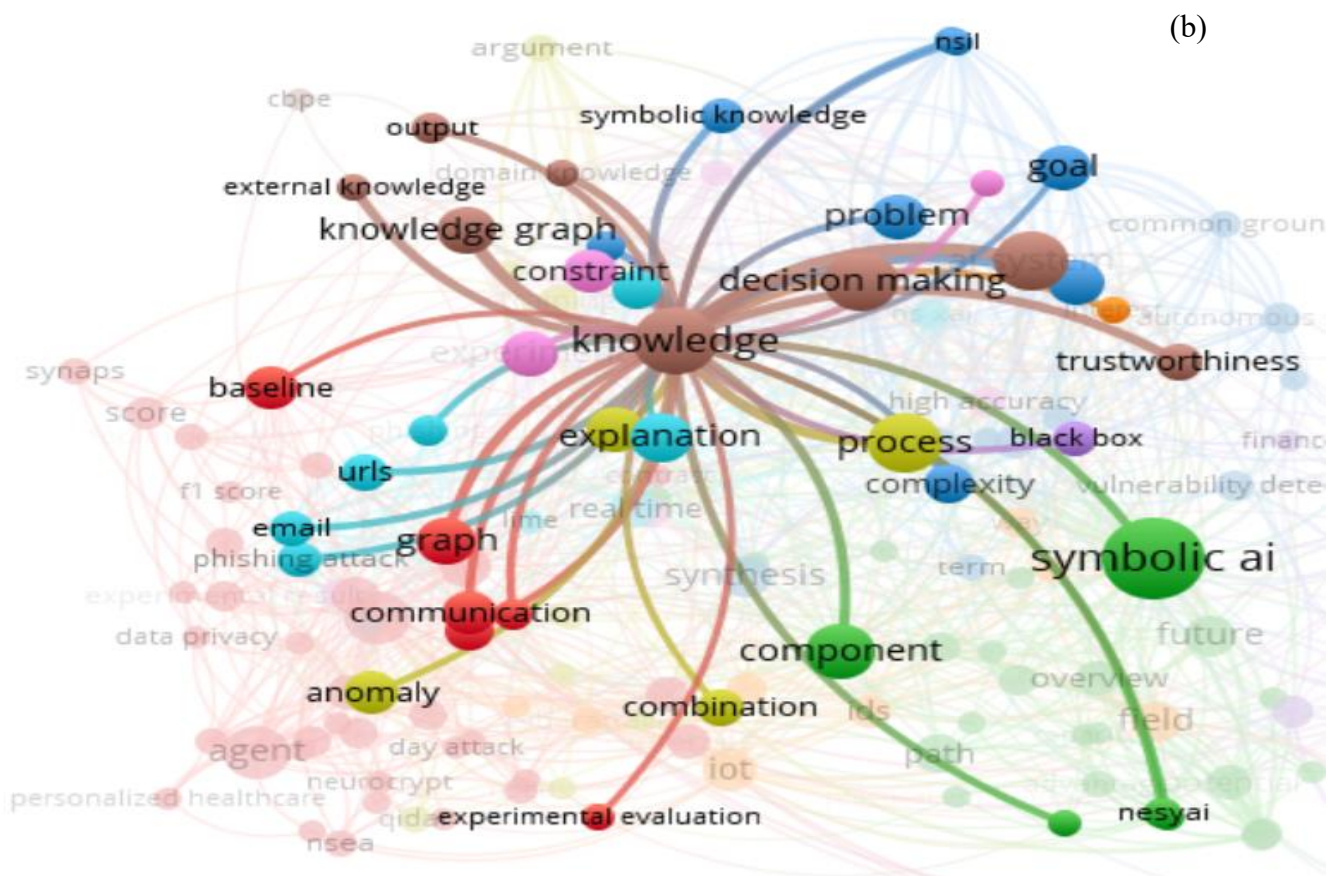
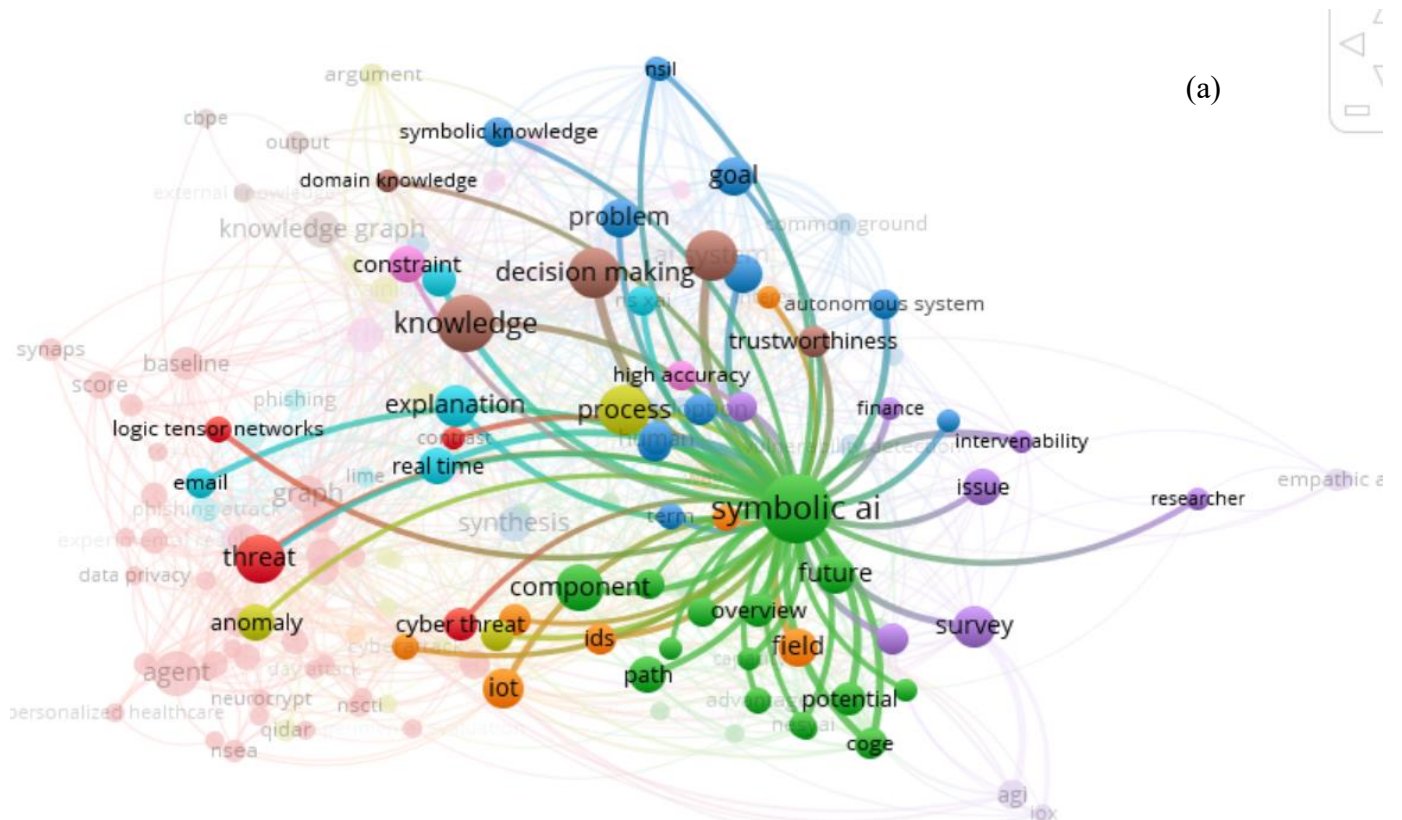


Figure 3. Foundations of Neuro-Symbolic AI

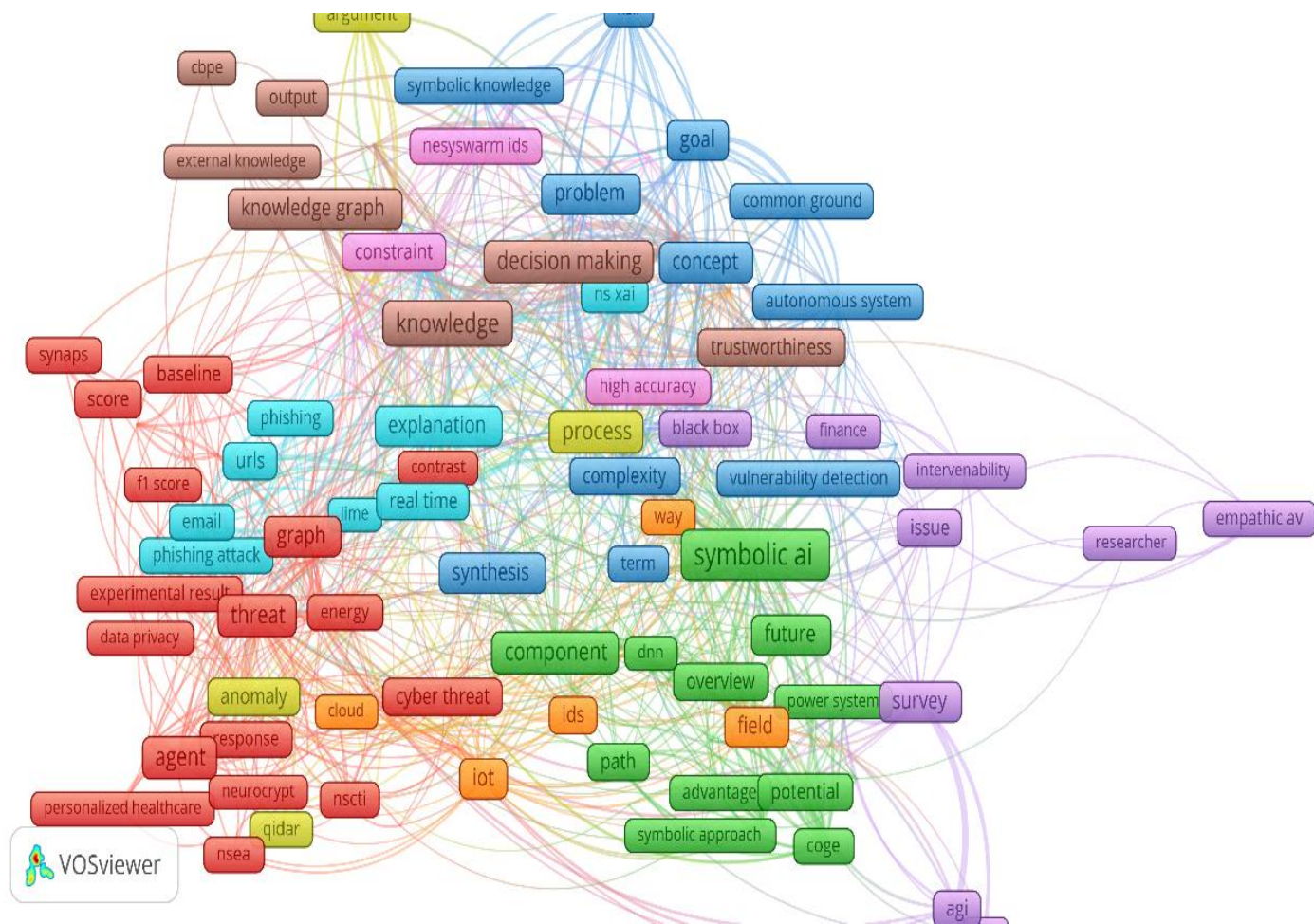


Figure 4- Deep neural network: input (teal), hidden layers (purple), output (coral)

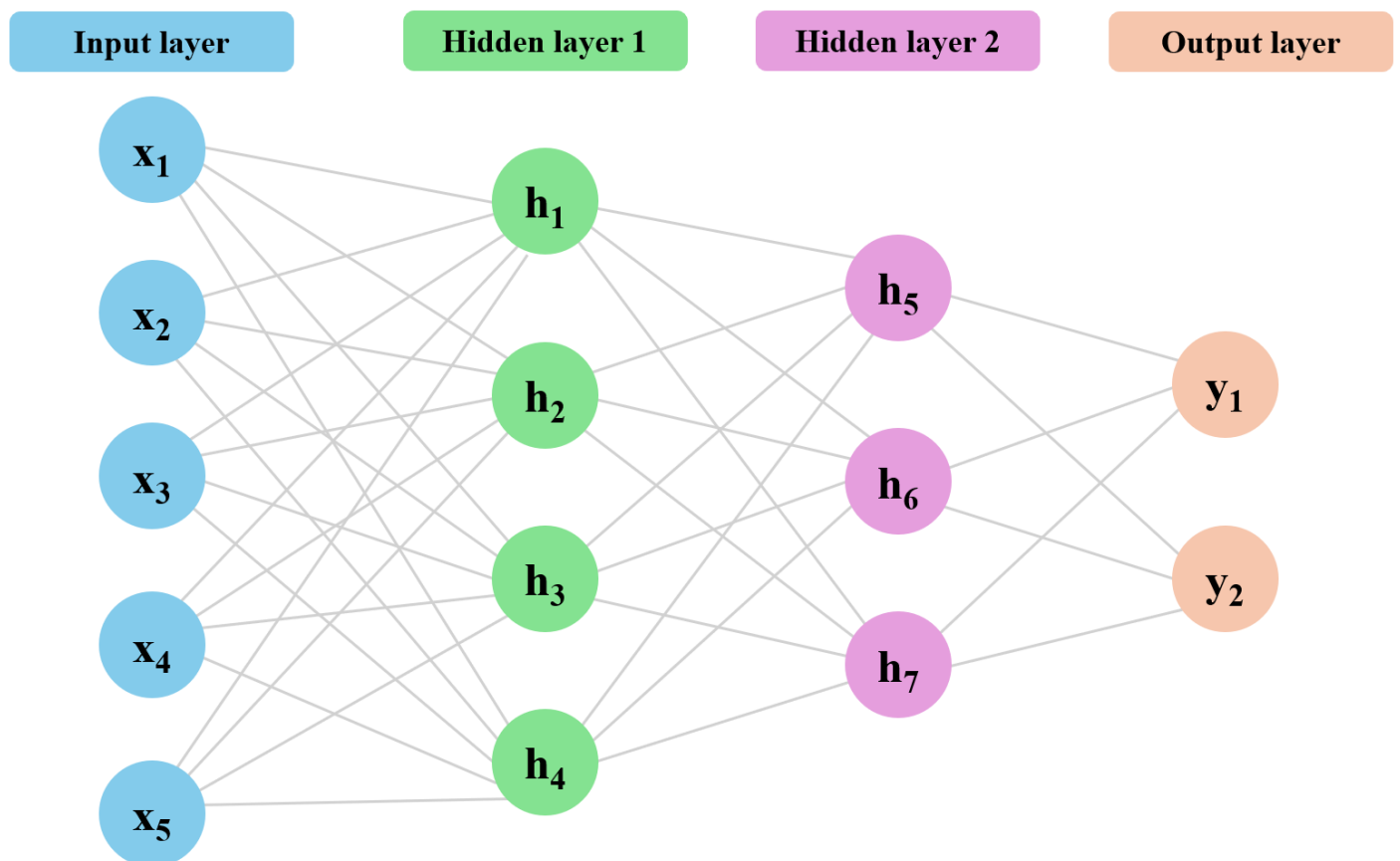


Figure. 5- Kautz's six-type neuro-symbolic taxonomy with cybersecurity examples

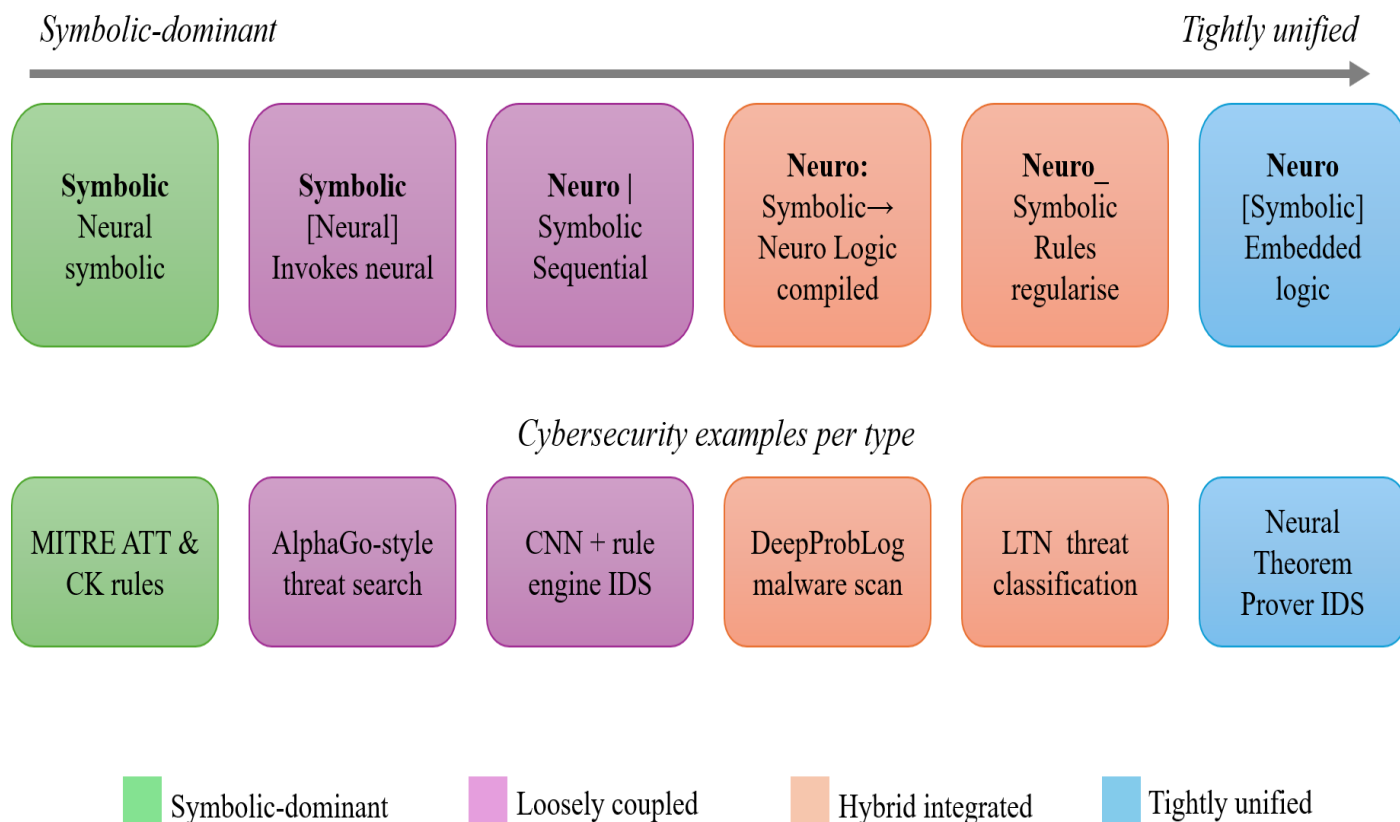


Figure 6: The four frameworks and their operational principles in a comparative structural layout

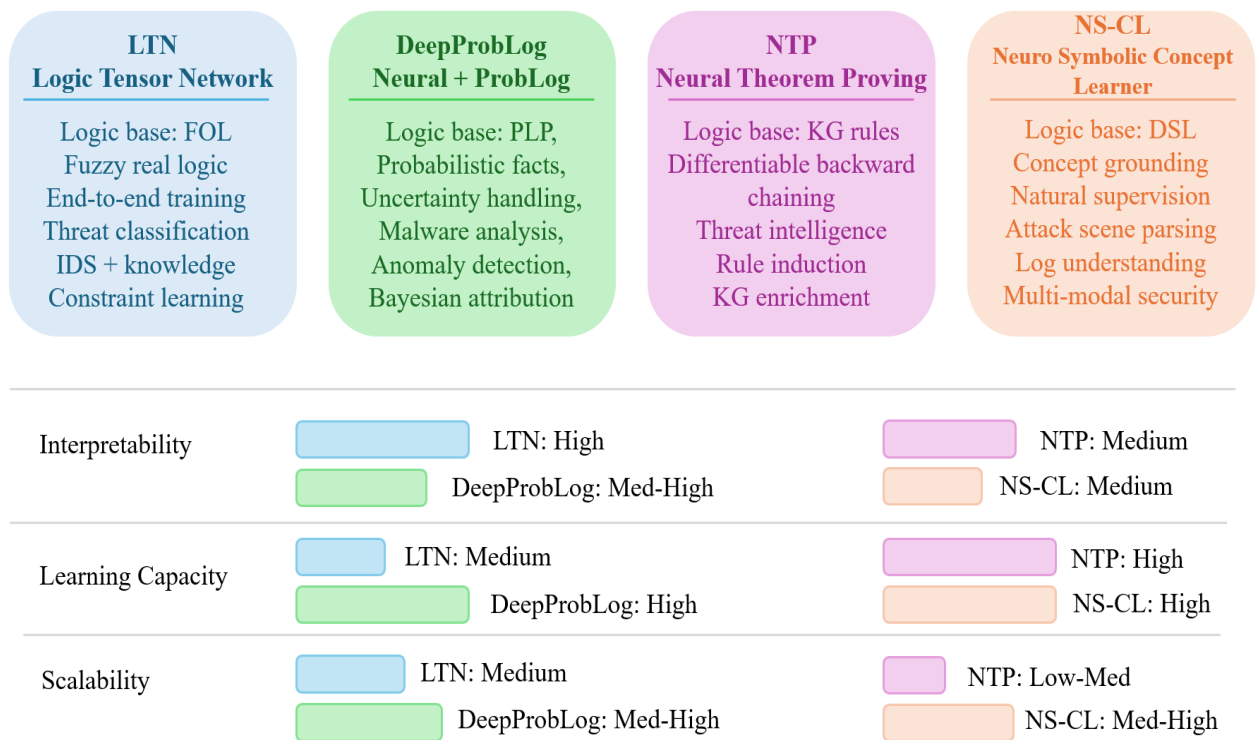


Figure. 7-End-to-end neuro-symbolic AI pipeline cybersecurity applications

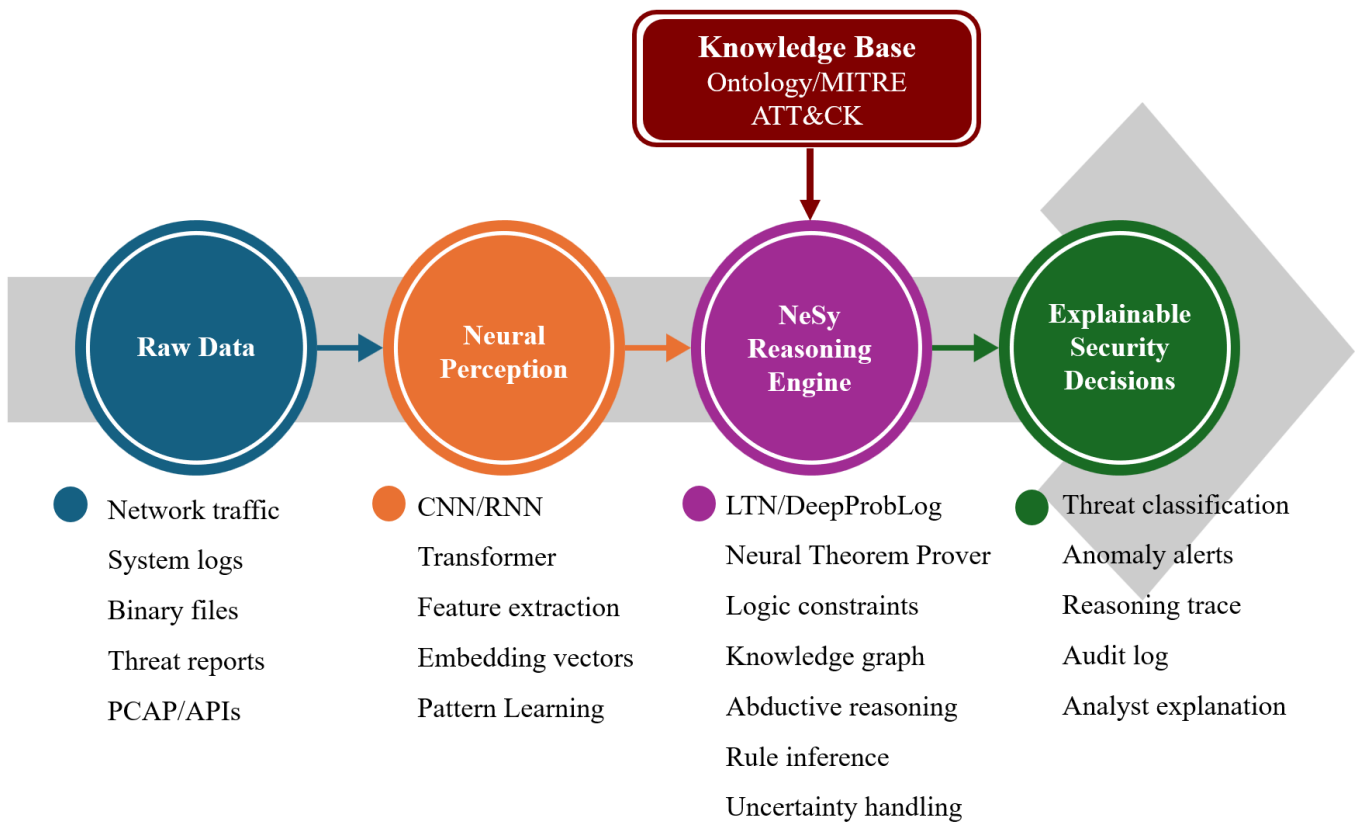


Figure. 8-Cybersecurity threat landscape statistics 2024-2025.

Global cybercrime cost
\$10.5T
projected 2025

Avg. data breach cost
\$4.88M
IBM 2024, +10% YoY

AI-driven attacks 2025
28M+
+72% year-over-year

Exploit window
22 min
CVE to weaponisation

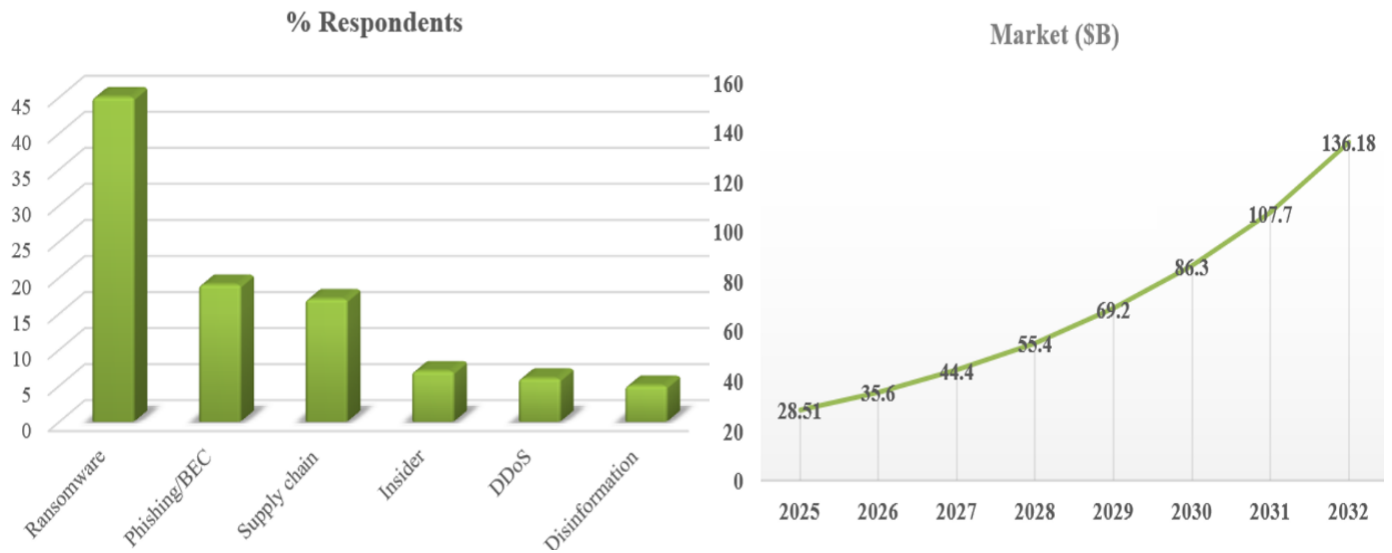


Figure 9: Four generations of AI in cybersecurity: capabilities and limitations at each stage

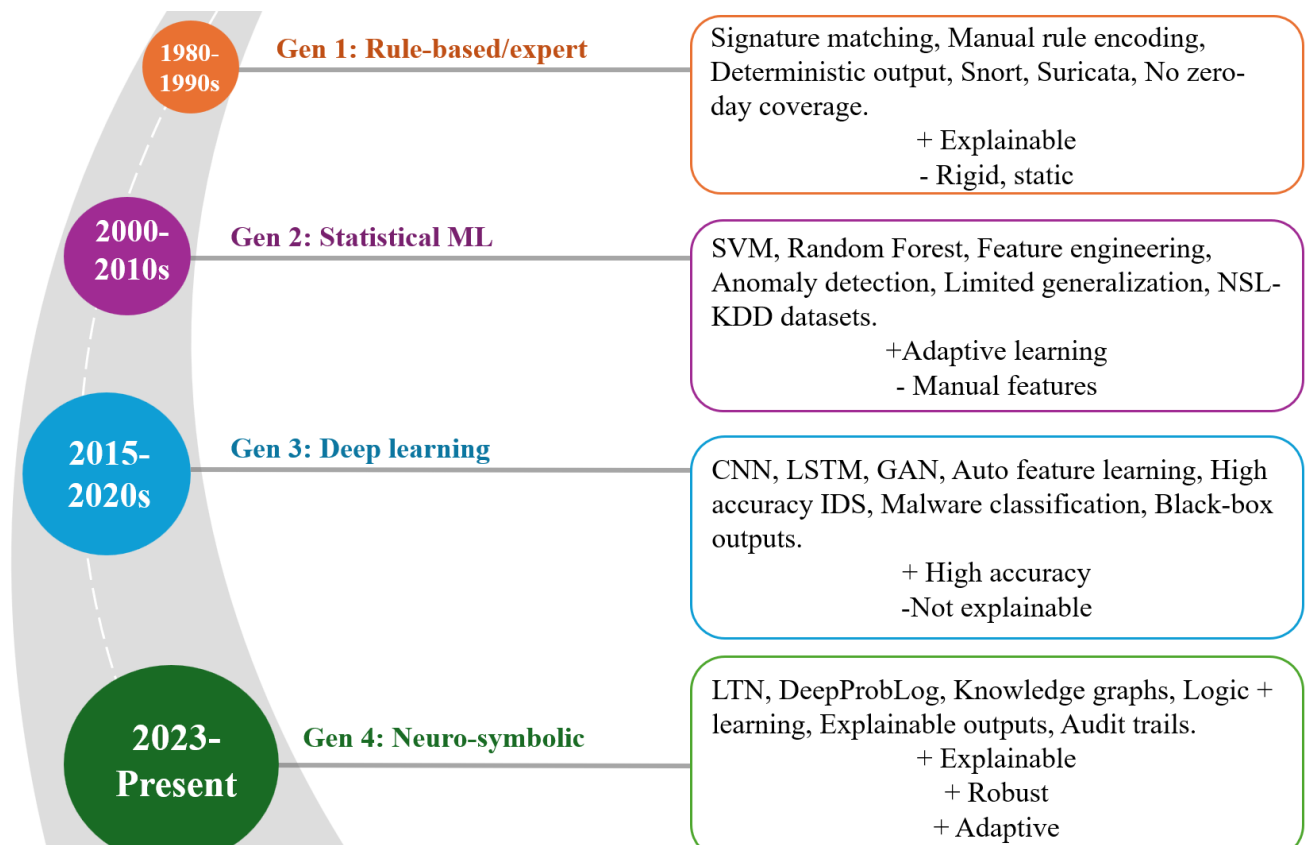


Figure. 10-Neuro-symbolic AI properties mapped to cybersecurity operational requirements

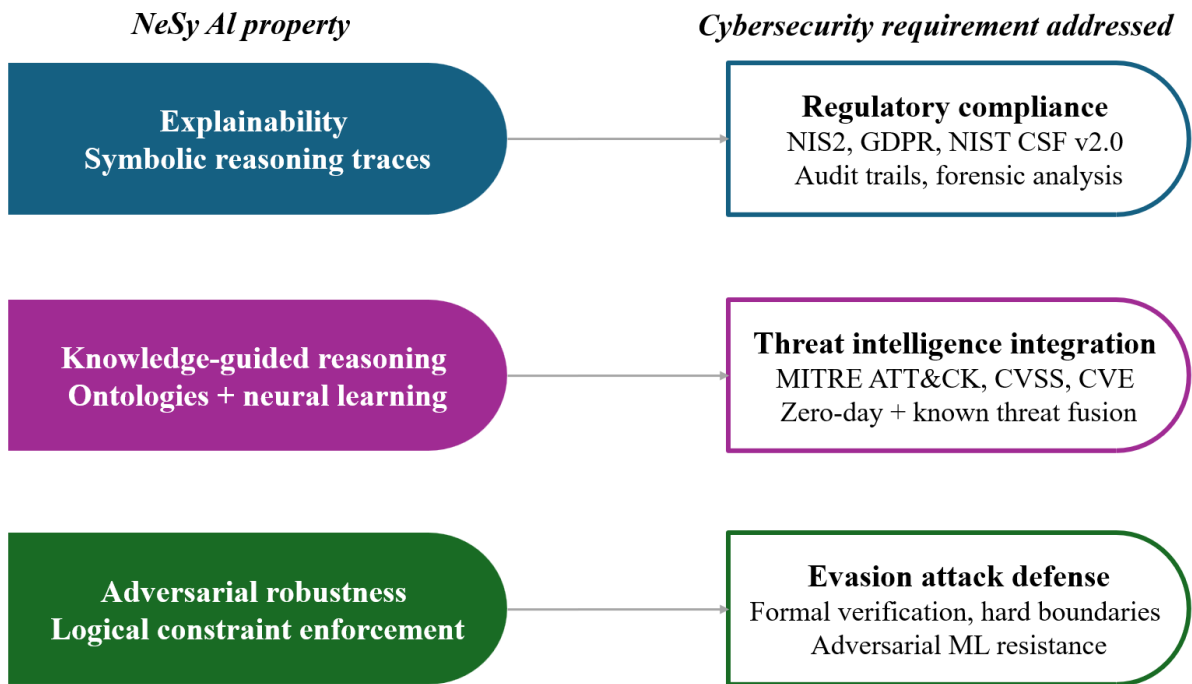


Figure. 11-Neuro-symbolic IDS architecture with CIA-triad validation and SOC feedback loop

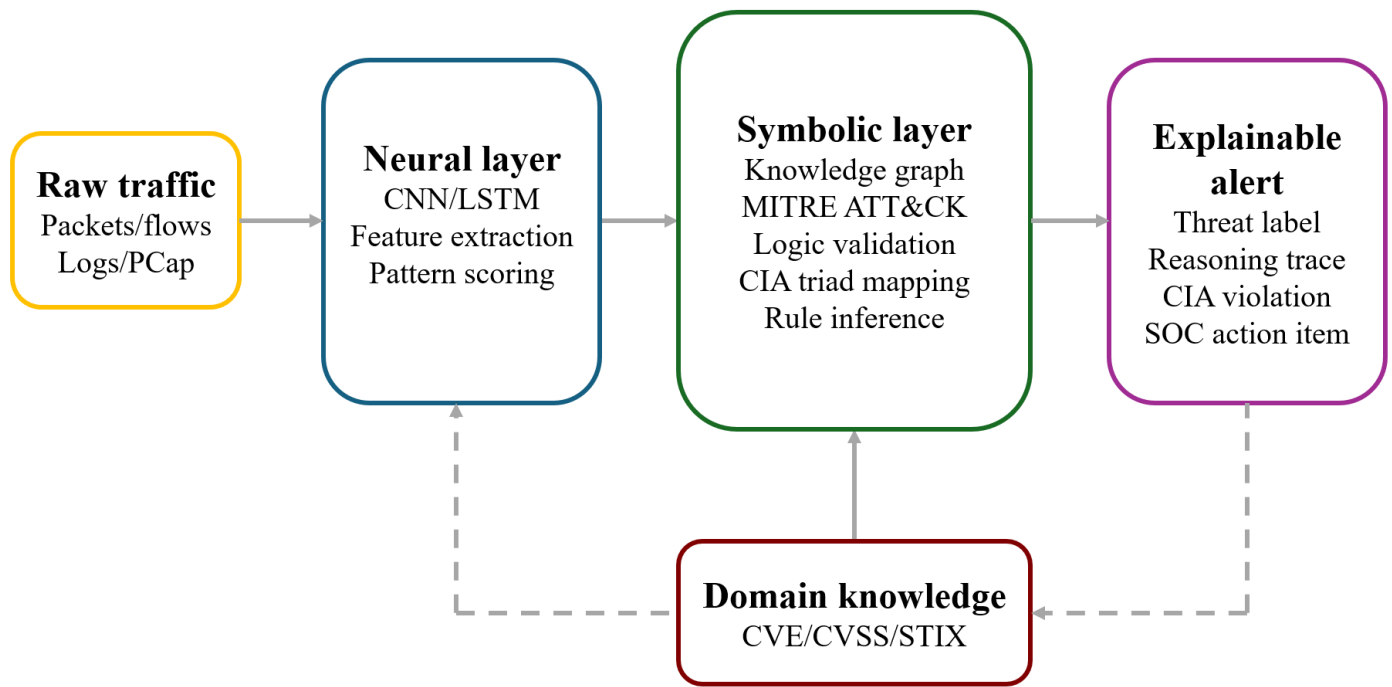


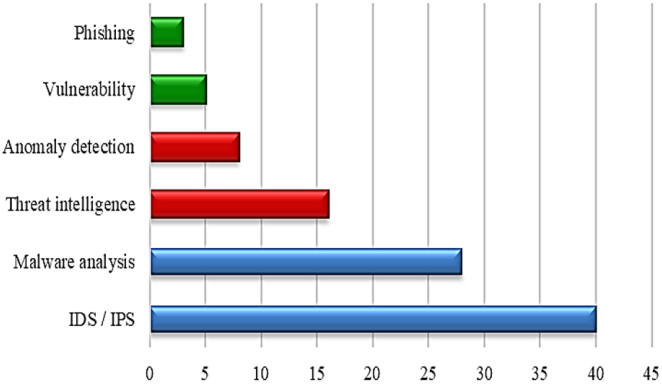
Figure. 12-NSAI application domain maturity and accuracy improvement over pure deep learning baselines.

Most mature domain
IDS / IPS
40%+ of NSAI publications

Fastest growing
Threat intelligence
LLM + KG fusion 2024–25

Highest accuracy gain in
Vulnerability assessment
+17.6% precision (MoCQ)

Share of publications (%) - Domain maturity level
(research publications, 2019–2025)



Accuracy improvement over pure DL (%) - NSAI vs
pure DL: performance improvement per domain (%)

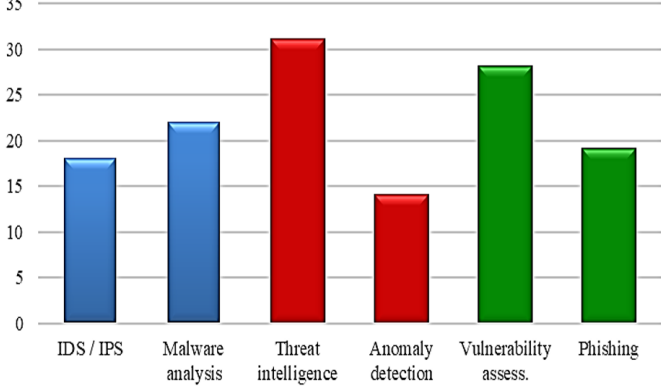


Figure. 13-Strategic positioning of NSAI application domains by knowledge richness and data availability

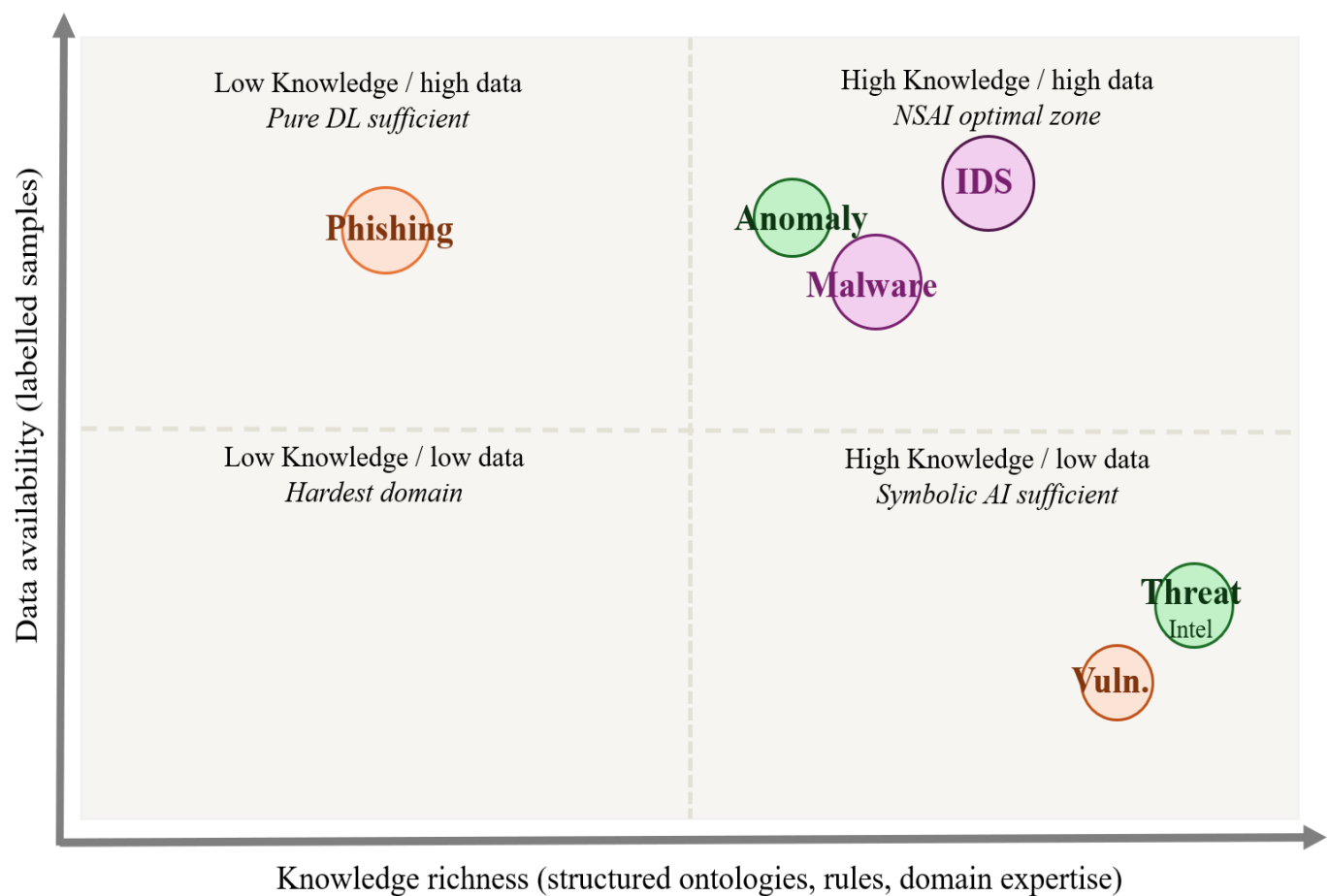


Figure 14 presents a clear table that lists the top cybersecurity datasets used for NSAI research, organized by threat domain, generation and basic attributes.

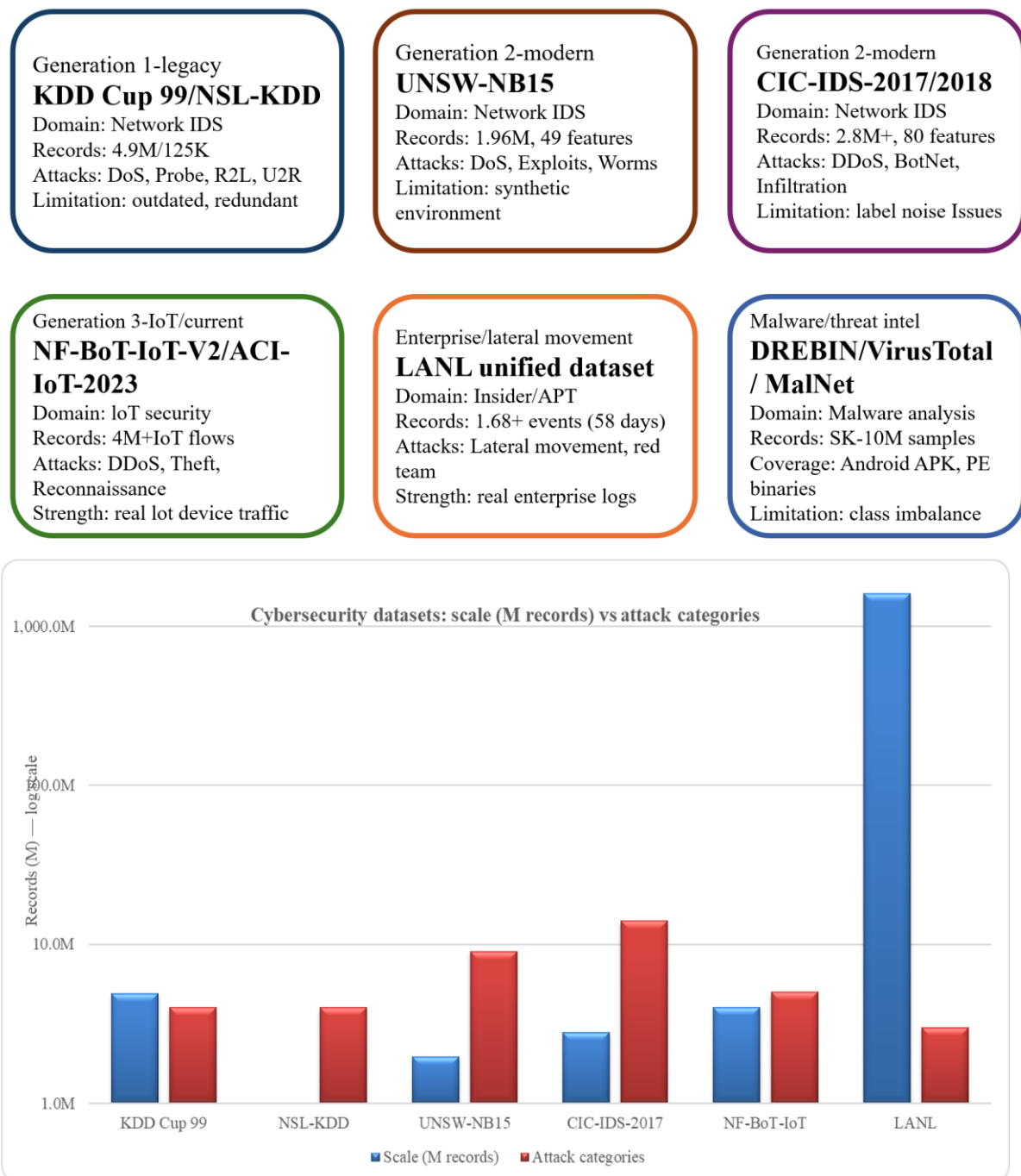


Figure 15: The three-dimensional performance space for the three competing paradigms across eight evaluation dimensions, using a grouped radar-style visualisation.

■ Neuro-symbolic AI ■ Pure deep learning ■ Rule-based / expert systems

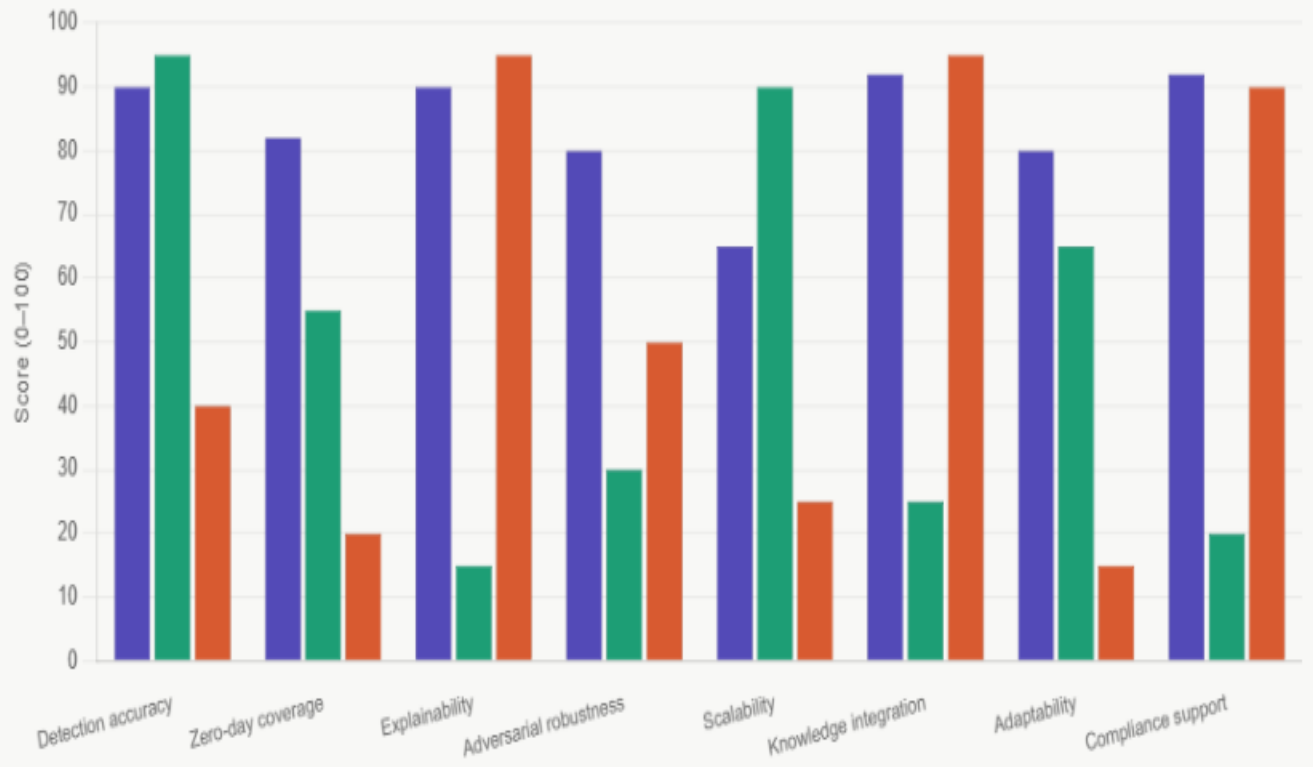


Table 1 — Comparative Limitations of Pure Neural and Symbolic Approaches in Cybersecurity (Acharya and Song, 2025; Antwi-Boasiako et al., 2024; Czaja et al., 2025; Kharchenko et al., 2022; Mohale and Obagbuwa, 2025; Ramachandram et al., 2025; Sharma et al., 2025)

Dimension	Pure Neural AI	Pure Symbolic AI
Interpretability	Low (black-box)	High (rule-transparent)
Scalability	High	Low
Adversarial Robustness	Vulnerable	Relatively rigid
Data Dependency	High	Low
Adaptability to New Threats	Moderate	Poor
Knowledge Encoding	Implicit (learned)	Explicit (manual)
Handling Unstructured Data	Strong	Weak
Formal Reasoning	Absent	Strong

Table 2 — Comparative analysis of cybersecurity defence paradigms across key evaluation dimensions [(Apruzzese et al., 2023; Reynaud and Roxin, 2025; Sarker et al., 2024; Tran et al., 2025)]

Evaluation dimension	Traditional/rule-based	Pure neural AI (DL/ML)	Neuro-symbolic AI
Threat detection scope	Signature-matched known threats only	Broad pattern learning, including anomalies	Broad learning + logical constraint enforcement
Zero-day attack handling	Poor — requires manual signature update	Moderate — anomaly detection possible	Good — symbolic constraints guide novel inference
Explainability	High-level — rules are human-readable	Low — black-box decisions	High — reasoning traces + logic audit trails
Adversarial robustness	Low — bypass via obfuscation	Low — susceptible to adversarial samples	Improved — symbolic constraints limit attack surface
Scalability	Poor — rule-based grows exponentially	High — scales with data volume	Moderate — depends on symbolic engine complexity
Knowledge integration	Manual and static	Implicit via training data	Explicit knowledge graphs + ontologies (e.g. MITRE ATT&CK)
Adaptability to new threats	Very low — requires expert re-authoring	Moderate — retraining required	Good — neural component adapts; symbolic component constrains
Regulatory compliance support	High — deterministic policy enforcement	Low — opacity limits audit	High — logical traces support GDPR/NIS2 compliance

Table 3: Cross-domain evaluation of neuro-symbolic AI applications in cybersecurity (2019–2025)

Application domain	NS integration paradigm	Representative frameworks/systems	Key datasets used	Key performance metrics	Maturity level
Intrusion detection (IDS/IPS)	Neural feature extraction + symbolic rule validation	NeSy-IDS, KiNetGAN, ZT-ID-IoV, dual-model CNN+ANN	NSL-KDD, UNSW-NB15, CIC-IDS-2017, NF-BoT-IoT-V2, ACI-IoT-2023	Accuracy >98%, F1 >0.97, FPR <0.5%, AUC >0.99	Mature
Malware detection and analysis	Knowledge graph embedding + neural classification	DeepProbLog-Malware, KG + DRL (Lin et al.), BERT-CKG	VirusTotal, DREBIN, MalNet, API call traces	Precision +17.6%, Recall +10%, 7 zero-days identified	Mature
Threat intelligence	LLM + knowledge graph symbolic reasoning	AGIR, LLM-KG threat hunting, STIX-CKG	STIX/TAXII feeds, CTI reports (474 reports corpus)	46 new patterns, 4 exclusive discoveries	Growing
Vulnerability assessment	Logic-constrained neural scoring	MoCQ, IRIS (LLM + static analysis), Licality-likelihood	NVD, CVE, CVSS, GitHub code repositories	+17.6% precision, +10% recall vs baseline	Emerging
Network anomaly detection	CIA-triad symbolic validation + neural anomaly scoring	NS-Anomaly (Clausen et al.), CBAM, Embed2Sym	LANL dataset, CAIDA, IoT-23	TP rate >97% at FPR 0.5%, AUC >0.98	Growing
Phishing / social engineering	Symbolic constraint enforcement + neural text/visual classifiers	NLP-AI phishing (Hussain et al.), SPF+neural fusion	PhishTank, OpenPhish, Email header datasets	F1 >0.95, evasion resistance +32% vs pure DL	Emerging

Table 4 — Recommended evaluation metric framework for neuro-symbolic AI in cybersecurity

Metric category	Specific metric	Definition	NSAI-specific relevance
Detection performance	Accuracy	$(TP + TN) / \text{Total}$	Baseline detection correctness
Detection performance	Precision	$TP / (TP + FP)$	Critical for minimising analyst alert fatigue
Detection performance	Recall (sensitivity)	$TP / (TP + FN)$	Critical for zero-day and novel attack coverage
Detection performance	F1-score	Harmonic mean of precision and recall	Balanced metric for imbalanced datasets
Detection performance	AUC-ROC	Area under the receiver operating curve	Overall discriminative power at all thresholds
Detection performance	FPR at fixed TPR	FP rate when TP rate is fixed (e.g. 0.5%)	Real-world SOC deployability criterion
Reasoning quality	Symbolic consistency	% of decisions consistent with the knowledge base	Verifies the symbolic layer is enforcing constraints
Reasoning quality	Explanation fidelity	Faithfulness of trace to actual decision path	Critical for audit and forensic use cases
Reasoning quality	Rule coverage	% of detections supported by symbolic rules	Measures the hybrid vs pure-neural reliance balance
Reasoning quality	Abduction accuracy	Correctness of inferred attack hypotheses	Specific to backwards-chaining NSAI systems
Operational efficiency	Inference latency (ms)	Time from input to decision output	Real-time deployment viability
Operational efficiency	Throughput (flows/sec)	Detection rate under peak traffic conditions	Enterprise scalability requirement
Operational efficiency	Data efficiency	Performance at reduced training set sizes	Measures the benefit of symbolic knowledge injection
Operational efficiency	Concept drift resistance	Performance decay rate over time without retraining	Measures the symbolic component's stabilising effect
Compliance suitability	Explainability score	Human expert rating of decision rationale quality	NIS2/GDPR audit trail requirement

Compliance suitability	Reproducibility index	Variance across independent re-runs	Benchmark standardisation requirement
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Table 5 — Comprehensive comparative analysis: neuro-symbolic AI vs pure deep learning vs rule-based systems in cybersecurity

Evaluation dimension	Pure deep learning	Rule-based/expert systems	Neuro-symbolic AI	Winner
Known threat detection accuracy	Very high (>99% on benchmarks)	High for signature-matched threats	High (>98%, logic-constrained)	Pure DL (in-distribution)
Zero-day / novel attack detection	Moderate (anomaly-based only)	Very low (signature-dependent)	High (symbolic inference + neural)	NSAI
Explainability and audit trails	Very low (black-box)	Very high (human-readable rules)	High (symbolic reasoning traces)	Rule-based / NSAI joint
Adversarial robustness	Low (evasion-susceptible)	Moderate (rigid, bypass via obfuscation)	Improved (constraint-bounded)	NSAI
Scalability to large networks	Very high	Low (exponential rule growth)	Moderate (knowledge graph scaling)	Pure DL
Knowledge and ontology integration	Implicit (training-encoded only)	Explicit and structured (manual)	Explicit + adaptive (KG + learning)	NSAI
Adaptability to concept drift	Moderate (retraining required)	Very low (expert re-authoring)	Good (neural adapts, symbolic constrains)	NSAI
Data efficiency	Low (large labelled sets required)	High (no data needed)	High (symbolic priors supplement data)	Rule-based / NSAI joint

Regulatory compliance (NIS2/GDPR)	Low (opacity issues)	High (deterministic and auditable)	High (symbolic traces + audit logs)	Rule-based / NSAI joint
False positive management	Moderate (high without tuning)	Moderate (rule-specificity limits FP)	Low (symbolic validation reduces FP)	NSAI
Real-time inference speed	Fast (GPU-accelerated)	Fast (lookup-based)	Moderate (symbolic overhead adds latency)	Pure DL / Rule-based
Integration with MITRE ATT&CK	Limited (requires training data)	Good (manual TTP encoding)	Excellent (KG-embedded ontology)	NSAI
Multi-domain threat reasoning	Poor (siloed models)	Poor (domain-specific rules only)	Good (shared symbolic ontology)	NSAI
Overall deployment readiness	Mature	Mature	Emerging–growing	Pure DL / Rule-based

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.